

ANALYSIS OF EM-DAT: AN R MINI PROJECT

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1. INTRODUCTION

EM-DAT was designed in 1988 based on an anthropocentric vision of disasters and emergencies. It considers disasters to be events involving an unexpected and overwhelming harmful impact on human beings. Formally, EM-DAT's definition of a disaster is:

A situation or event which overwhelms local capacity, necessitating a request to the national or international level for external assistance; an unforeseen and often sudden event that causes great damage, destruction, and human suffering.

For more information, kindly visit: [EM-DAT DATABASE](#)

This analysis aims to explore the global landscape of disasters as recorded in the EM-DAT database, with a focus on understanding the timeline and evolution of disaster occurrences, the classification of disasters, and the identification of the most affected regions worldwide.

In addition to general trends, special attention is given to case studies that provide deeper insights into specific events and regions. These include a focused analysis of the 2004 Indian Ocean Earthquake and Tsunami and an overview of disasters in India.

2. OBJECTIVES

This analysis is primarily exploratory in nature, with the following objectives carefully formulated after a thorough examination of the dataset.

A. One-Dimensional Summaries

These univariate analyses explore the frequency and duration of disasters across categories, year and regions individually.

- Analysis by Disaster Classification
- Analysis by Region
- Analysis by Disaster Start Year

B. Two-Dimensional Summaries

These two-dimensional summaries explore relationships between disaster characteristics such as classification, region, and start year to uncover patterns and trends across the dataset.

- Analysis by Disaster Classification and Region
- Analysis by Disaster Start Year and Region

C. TopN Deep-Dives

This section presents focused deep-dive analyses into the top contributors to total deaths, examined by disaster classification, region, and start year to identify the most impactful events and trends.

- Analysis of Total Deaths by Disaster Classification
- Analysis of Total Deaths by Disaster Start Year

D. Case Studies

This section presents detailed case studies, including the 2004 Indian Ocean tsunami, an in-depth disaster analysis of India.

- Disasters in India
- The 2004 Indian Ocean Earthquake and Tsunami

3. PRELIMINARY ANALYSIS

The preliminary section involves setting up the environment by installing libraries, loading datasets, and performing essential preprocessing tasks such as column selection, handling missing and duplicate values, and basic feature engineering.

3.1 Installing and Attaching Packages

Installing and attaching necessary packages to be used throughout the analysis.

```
# Specify a CRAN mirror before installing the package
#options(repos = c(CRAN = "https://cran.rstudio.com/"))
#install.packages('tidyr')
#install.packages('dplyr')
#install.packages('ggplot2')
#install.packages('lubridate')
#install.packages('stringr')
#install.packages('readr')
#install.packages('knitr')

# Attaching installed packages
library('tidyr')
library('dplyr')
library('ggplot2')
library('lubridate')
library('stringr')
library('readr')
library('knitr')
```

3.2 Importing Datasets

Importing datasets from local working directory

```
# Setting the path to access file locations
path <- "~/Mini Project/"
for (i in c('Asia', 'Europe', 'Oceania', 'Africa', 'America'))
{
  file_path <- paste(path, i, '.csv', sep = "")
  temp_data <- read.csv(file_path)
  df_name <- paste(i, "_df", sep="")
  assign(df_name, temp_data)
}
```

3.3 Data Preprocessing

Merging the datasets together and taking a glimpse at the combined data

```
# Checking if all the datasets have same columns to ease up concatenation of dfs
as <- as.vector(names(Asia_df))
eu <- as.vector(names(Europe_df))
```

```

oc <- as.vector(names(Oceania_df))
af <- as.vector(names(Africa_df))
am <- as.vector(names(America_df))

all_cols <- unique(c(as, eu, oc, af, am))

datasets <- list(Asia = as, Europe = eu, Oceania = oc, Africa = af, America = am)
names(datasets)

```

```
## [1] "Asia"      "Europe"    "Oceania"  "Africa"   "America"
```

```

for (name in names(datasets))
{
  dataset_columns <- datasets[[name]]
  missing_cols <- setdiff(all_cols, dataset_columns)
  if (length(missing_cols)>0)
  {
    print(paste("Dataset", name, "doesn't have the following columns:"))
    print(missing_cols)
  }
  else
  {
    print(paste("Dataset", name, "has all the columns."))
  }
}

```

```

## [1] "Dataset Asia has all the columns."
## [1] "Dataset Europe has all the columns."
## [1] "Dataset Oceania has all the columns."
## [1] "Dataset Africa has all the columns."
## [1] "Dataset America has all the columns."

```

```

# Since all the datasets have same columns, we'll proceed with appending rows
df <- bind_rows(Asia_df, America_df, Africa_df, Europe_df, Oceania_df)

# Cleaning column names
# Uppercasing column names, substituting blanks with _ & trimming leading spaces
names(df) <- toupper(str_replace_all
  (str_replace_all(str_trim(names(df))," ","_"),"\\.", "_"))
glimpse(df)

```

```

## Rows: 16,194
## Columns: 46
## $ DISNO_                <chr> "2000-0003-BGD", "2000-0005--
## $ HISTORIC              <chr> "No", "No", "No", "No", "No~
## $ CLASSIFICATION_KEY    <chr> "nat-met-ext-col", "tec-tra~
## $ DISASTER_GROUP        <chr> "Natural", "Technological", ~
## $ DISASTER_SUBGROUP     <chr> "Meteorological", "Transport~
## $ DISASTER_TYPE         <chr> "Extreme temperature", "Road~
## $ DISASTER_SUBTYPE      <chr> "Cold wave", "Road", "Fire (~
## $ EXTERNAL_IDS          <chr> "", "", "", "", "", "", "", ~
## $ EVENT_NAME            <chr> "", "", "Hotel", "", "", "", ~

```

```
## $ ISO <chr> "BGD", "CHN", "CHN", "IRN", ~
## $ COUNTRY <chr> "Bangladesh", "China", "Chin~
## $ SUBREGION <chr> "Southern Asia", "Eastern As~
## $ REGION <chr> "Asia", "Asia", "Asia", "Asi~
## $ LOCATION <chr> "Barisal, Chittagong, Dhaka,~
## $ ORIGIN <chr> "", "", "", "", "", "", "", ~
## $ ASSOCIATED_TYPES <chr> "", "", "", "Fire", "", "", ~
## $ OFDA_BHA_RESPONSE <chr> "No", "No", "No", "No", "No"~
## $ APPEAL <chr> "No", "No", "No", "No", "No"~
## $ DECLARATION <chr> "No", "No", "No", "No", "No"~
## $ AID_CONTRIBUTION___000_US__ <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ MAGNITUDE <dbl> 6.4, NA, NA, 120.0, NA, NA, ~
## $ MAGNITUDE_SCALE <chr> "°C", "", "", "Kph", "", "", ~
## $ LATITUDE <dbl> NA, NA, NA, NA, NA, NA, NA, ~
## $ LONGITUDE <dbl> NA, NA, NA, NA, NA, NA, NA, ~
## $ RIVER_BASIN <chr> "", "", "", "", "", "", "", ~
## $ START_YEAR <int> 2000, 2000, 2000, 2000, 2000~
## $ START_MONTH <int> 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ START_DAY <int> NA, 1, 9, 18, 5, 3, NA, 2, 2~
## $ END_YEAR <int> 2000, 2000, 2000, 2000, 2000~
## $ END_MONTH <int> 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ END_DAY <int> NA, 1, 9, 18, 5, 3, NA, 2, 2~
## $ TOTAL_DEATHS <int> 49, 22, 12, 3, 13, 13, 275, ~
## $ NO__INJURED <int> NA, 29, 18, NA, 22, NA, NA, ~
## $ NO__AFFECTED <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ NO__HOMELESS <int> NA, NA, NA, 5500, NA, NA, NA~
## $ TOTAL_AFFECTED <int> NA, 29, 18, 5500, 22, NA, NA~
## $ RECONSTRUCTION_COSTS___000_US__ <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ RECONSTRUCTION_COSTS__ADJUSTED___000_US__ <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ INSURED_DAMAGE___000_US__ <dbl> NA, NA, NA, NA, NA, NA, NA, ~
## $ INSURED_DAMAGE__ADJUSTED___000_US__ <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ TOTAL_DAMAGE___000_US__ <dbl> NA, NA, NA, NA, NA, NA, NA, ~
## $ TOTAL_DAMAGE__ADJUSTED___000_US__ <int> NA, NA, NA, NA, NA, NA, NA, ~
## $ CPI <dbl> 56.51429, 56.51429, 56.51429~
## $ ADMIN_UNITS <chr> "[{\\"adm1_code\\":575,\"adm1_~
## $ ENTRY_DATE <chr> "2003-07-01", "2005-02-09", ~
## $ LAST_UPDATE <chr> "2023-09-25", "2023-09-25", ~
```

```
# there are lot of NA's in the dataset.
# This can be a painful point in the analysis going ahead.
```

```
# Peeking at the nulls and NA's and blanks in the data
missing_df <- df %>%
  summarise(across(
    everything(),
    ~ mean(is.na(.) | (. == "" & is.character(.))) * 100
  )) %>%
  pivot_longer(everything(), names_to = 'COLUMN', values_to = 'MISSING_PCT')

missing_df %>% arrange(desc(MISSING_PCT))
```

```
## # A tibble: 46 x 2
##   COLUMN MISSING_PCT
```

```
##      <chr>                                <dbl>
## 1 RECONSTRUCTION_COSTS___000_US__         99.8
## 2 RECONSTRUCTION_COSTS__ADJUSTED___000_US__ 99.8
## 3 AID_CONTRIBUTION___000_US__             97.0
## 4 INSURED_DAMAGE__ADJUSTED___000_US__      95.7
## 5 INSURED_DAMAGE___000_US__               95.6
## 6 RIVER_BASIN                             92.4
## 7 NO__HOMELESS                           91.8
## 8 LATITUDE                                88.8
## 9 LONGITUDE                               88.8
## 10 EXTERNAL_IDS                           83.3
## # i 36 more rows
```

```
# there are a handful of columns which have near to 100% missing rows pct
# These columns would be removed in column removal section
```

3.3.1 Preprocessing Parameters

For further preprocessing, creating a list of preprocessing parameters. This will enable quick and efficient preprocessing of the datasets in one go.

```
# Scope and Feature Selection
col_selection <- FALSE
cols_to_select <- c()
col_removal <- TRUE
cols_to_remove <- c("HISTORIC", "EXTERNAL_IDS", "EVENT_NAME", "ADMIN_UNITS",
                    "ENTRY_DATE", "LAST_UPDATE", "LOCATION", "ORIGIN",
                    "RIVER_BASIN")

# location, origin, river basin are text heavy and would require NLP libraries.
# Besides the data is too less in these columns
# historic has only 1 value, external id, event name and admin id
# have too many missings; entry date & last update aren't required for analysis

## Data Cleaning
drop_duplicates <- TRUE
drop_nas <- TRUE
primary_key <- 'DISNO_'
```

3.3.2 Scope and Feature Selection

Filtering scope of the dataset wrt START_YEAR

```
df <- df %>% filter("START_YEAR">=2000)
print(paste("In scope dataset has", nrow(df), "rows."))
```

```
## [1] "In scope dataset has 16194 rows."
```

```
## 3.4 Feature Selection
if (col_selection==TRUE)
{
  df <- df %>% select(all_of(cols_to_select))
}
```

```

}
if (col_removal==TRUE)
{
  df <- df %>% select(-all_of(cols_to_remove))
}

print(paste("Dataset has ", length(df), "columns."))

```

```
## [1] "Dataset has 37 columns."
```

```
print(names(df))
```

```

## [1] "DISNO_"
## [2] "CLASSIFICATION_KEY"
## [3] "DISASTER_GROUP"
## [4] "DISASTER_SUBGROUP"
## [5] "DISASTER_TYPE"
## [6] "DISASTER_SUBTYPE"
## [7] "ISO"
## [8] "COUNTRY"
## [9] "SUBREGION"
## [10] "REGION"
## [11] "ASSOCIATED_TYPES"
## [12] "OFDA_BHA_RESPONSE"
## [13] "APPEAL"
## [14] "DECLARATION"
## [15] "AID_CONTRIBUTION___000_US_"
## [16] "MAGNITUDE"
## [17] "MAGNITUDE_SCALE"
## [18] "LATITUDE"
## [19] "LONGITUDE"
## [20] "START_YEAR"
## [21] "START_MONTH"
## [22] "START_DAY"
## [23] "END_YEAR"
## [24] "END_MONTH"
## [25] "END_DAY"
## [26] "TOTAL_DEATHS"
## [27] "NO__INJURED"
## [28] "NO__AFFECTED"
## [29] "NO__HOMELESS"
## [30] "TOTAL_AFFECTED"
## [31] "RECONSTRUCTION_COSTS___000_US_"
## [32] "RECONSTRUCTION_COSTS__ADJUSTED___000_US_"
## [33] "INSURED_DAMAGE___000_US_"
## [34] "INSURED_DAMAGE__ADJUSTED___000_US_"
## [35] "TOTAL_DAMAGE___000_US_"
## [36] "TOTAL_DAMAGE__ADJUSTED___000_US_"
## [37] "CPI"

```

3.3.3 Data Cleaning

Since, majority of the analysis scope revolves around data exploration, a basic level of data cleaning is done including removal of duplicates and NA's in the primary key of the dataset. Capping of outliers, missing value imputation, etc aren't followed to preserve the actual picture of the data.

For example, the missing aid values in the dataset appear to be *Missing Not At Random (MNAR)*, as the absence of data may reflect instances where aid was not given or was insufficiently recorded. Consequently, imputing these values with measures like the mean would introduce bias and distort the true distribution, so such values are left as missing or handled with caution in analyses.

Even the EM-DAT website claims: *the dataset is recorded as is.*

```
# Duplicates Check: there shall be no duplicates in the dataset wrt primary key
if (drop_duplicates == TRUE)
{
  l_df <- nrow(df)
  l_pk_df <- nrow(df %>% distinct(get(primary_key), .keep_all = TRUE))
  ##ensures all columns are kept, but unique is checked on primary_key only
  if( l_df == l_pk_df )
  {
    print("There are no dups in primary key of the dataset. You can proceed!")
  }
  else
  {
    print(paste("Either",primary_key,"is not the primary key of the dataset,
                or the dataset has duplicates"))
  }
}
```

```
## [1] "There are no dups in primary key of the dataset. You can proceed!"
```

```
# NA Check: there shall be no NAs in the dataset wrt primary key
if (drop_nas) {
  l_df <- nrow(df)
  l_pk_df <- nrow(df %>% drop_na(all_of(primary_key)))

  if (l_df == l_pk_df) {
    print("There are no NA's in primary key of the dataset. You can proceed!")
  } else {
    print("There are NA's in the primary key. Kindly, remove them.")
  }
}
```

```
## [1] "There are no NA's in primary key of the dataset. You can proceed!"
```

3.3.4 Feature Engineering

The start date and end date of a disaster are captured in days, months and years columns separately. There are a few missing in those columns as well. Creating a new column as START_DATE and END_DATE by unifying the individual components, and padding the NA's with '01' in day and month columns. This enables assumption that wherever a NA was present in the day column, the event happened on 1st day.

Also, calculating the duration of disaster as END_DATE - START DATE, and adding a flag to compute disaster duration only for valid cases.


```

# Padding nulls and unifying components to create a single date column
df <- df %>% mutate(START_DAY_ = ifelse(is.na(START_DAY),1,START_DAY),
                        START_MONTH_ = ifelse(is.na(START_MONTH),1,START_MONTH),
                        START_DATE= ymd(paste(START_YEAR, START_MONTH_, START_DAY_)),
                        END_DAY_ = ifelse(is.na(END_DAY),1,END_DAY),
                        END_MONTH_ = ifelse(is.na(END_MONTH),1,END_MONTH),
                        END_DATE= ymd(paste(END_YEAR, END_MONTH_, END_DAY_)) ) %>%
select(-c('START_DAY_', 'START_MONTH_', 'END_DAY_', 'END_MONTH_', 'START_DAY',
          'START_MONTH', 'END_DAY', 'END_MONTH'))

# Due to missing info in the date columns, the disaster duration calculation
# becomes messy. Hence, selecting only those cases, where start day < end date
df = df %>% mutate(DISASTER_DURATION = ifelse((END_DATE-START_DATE)<0,0,
                                                END_DATE-START_DATE)) %>%
mutate(DURATION_FLAG = ifelse(DISASTER_DURATION>0,1,0))

```

Now that preliminary data preprocessing is done, proceeding ahead with the detailed analysis of the data.

4. DETAILED ANALYSIS

The detailed analysis section will cover the objectives as listed under the relevant section of the document.

A. One-Dimensional Summaries

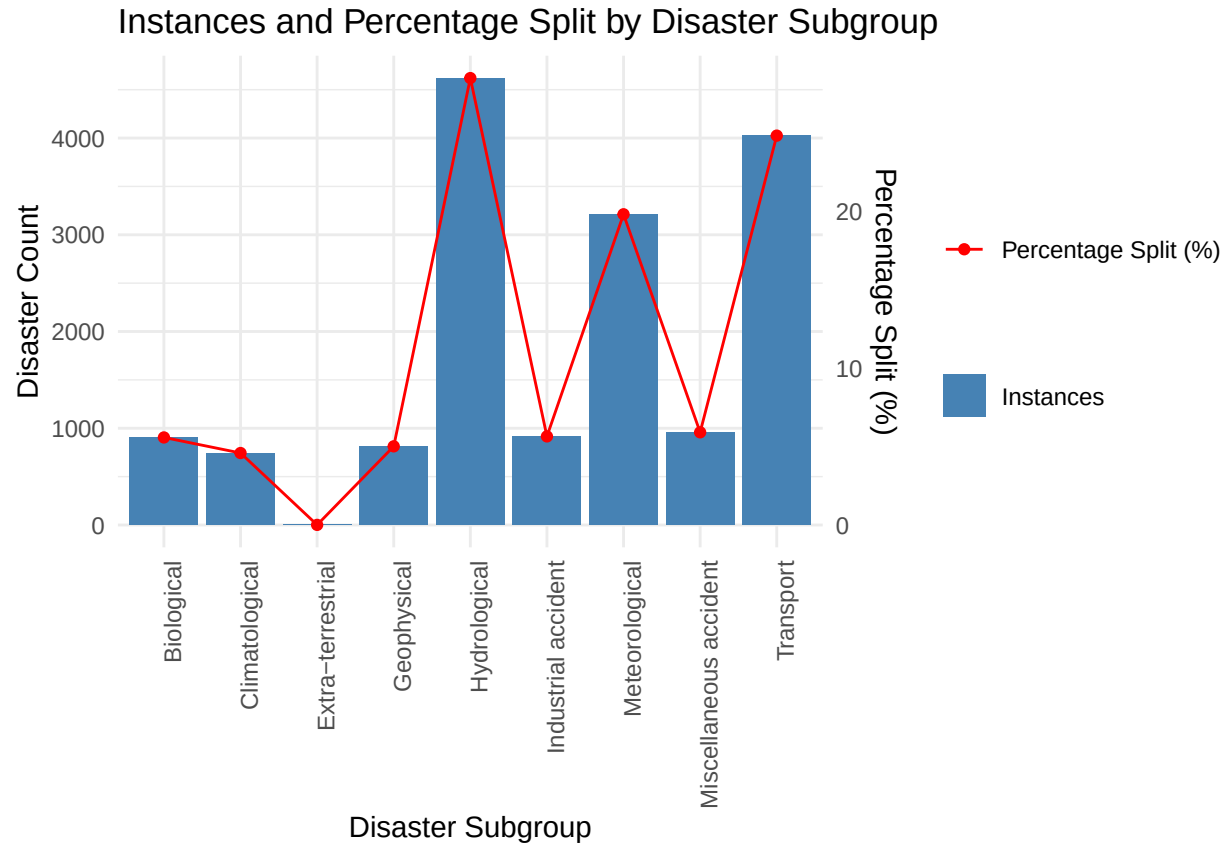
These univariate analyses explore the frequency and duration of disasters across categories, year and regions individually.

A.1 Analysis by Disaster Classification

- 1) Understanding the disaster instances by subgroups and subtypes

```
temp_df <- df %>% group_by(DISASTER_GROUP ,DISASTER_SUBGROUP) %>%
  summarize(INSTANCES = n(),PCT_SPLIT = n()/nrow(df)*100)

scale_factor <- max(temp_df$INSTANCES) / max(temp_df$PCT_SPLIT)
ggplot(temp_df, aes(x = DISASTER_SUBGROUP)) +
  geom_bar(aes(y = INSTANCES, fill = "Instances"), stat = "identity") +
  geom_line(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)",
               group = 1) +
  geom_point(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)"))+
  scale_y_continuous(
    name = "Disaster Count",
    sec.axis = sec_axis(~ . / scale_factor, name = "Percentage Split (%)")
  ) +
  scale_fill_manual(name = "", values = c("Instances" = "steelblue")) +
  scale_color_manual(name = "", values = c("Percentage Split (%)" = "red")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Instances and Percentage Split by Disaster Subgroup",
    x = "Disaster Subgroup"
  )
```



Since, subgroup ‘Hydrological’ has the highest peak in the bar chart, we’ll drill down a level to disaster subtype to understand the major disaster type.

```
temp_df <- df %>% group_by(DISASTER_GROUP ,DISASTER_SUBGROUP,
                           DISASTER_TYPE) %>%
  summarize(INSTANCES = n()) %>%
  arrange(desc(INSTANCES))
head(temp_df,10)
```

```
## # A tibble: 10 x 4
## # Groups:   DISASTER_GROUP, DISASTER_SUBGROUP [7]
##   DISASTER_GROUP DISASTER_SUBGROUP DISASTER_TYPE INSTANCES
##   <chr>          <chr>          <chr>      <int>
## 1 Natural      Hydrological    Flood      4140
## 2 Natural      Meteorological  Storm      2677
## 3 Technological Transport      Road       2188
## 4 Technological Transport      Water      1149
## 5 Natural      Biological      Epidemic    875
## 6 Natural      Geophysical     Earthquake  670
## 7 Natural      Meteorological  Extreme temperature  535
## 8 Natural      Hydrological    Mass movement (wet)  480
## 9 Technological Industrial accident Explosion (Industrial)  464
## 10 Technological Miscellaneous accident Fire (Miscellaneous)  439
```

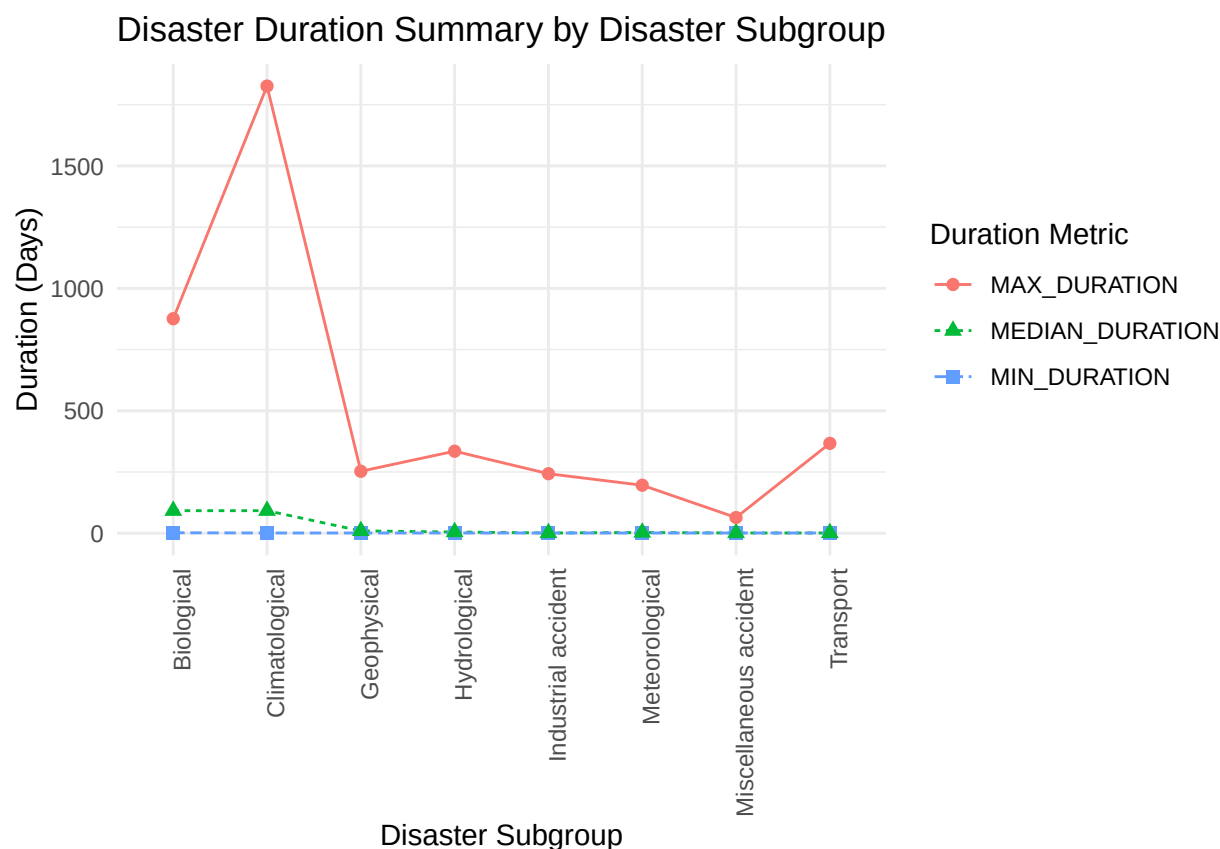
Clearly, the disaster type ‘Flood’ accounts for 4140 instances out of 4620 occurring under the disaster subgroup ‘Hydrological’. A detailed deep dive into the disaster ‘Flood’ will be carried out in the later sections.

2) Understanding the disaster duration by subgroups and subtypes

```
temp_df <- df %>% filter(DURATION_FLAG == 1) %>%  
  group_by(DISASTER_GROUP, DISASTER_SUBGROUP) %>%  
  summarize(MIN_DURATION = min(DISASTER_DURATION),  
            MEDIAN_DURATION = median(DISASTER_DURATION),  
            MAX_DURATION = max(DISASTER_DURATION))  
# considering only valid cases for disaster duration cases  
head(temp_df %>%  
  select(c(DISASTER_GROUP, DISASTER_SUBGROUP,  
           MEDIAN_DURATION, MAX_DURATION)),10)
```

```
## # A tibble: 8 x 4  
## # Groups:   DISASTER_GROUP [2]  
##   DISASTER_GROUP DISASTER_SUBGROUP MEDIAN_DURATION MAX_DURATION  
##   <chr>          <chr>          <dbl>         <dbl>  
## 1 Natural      Biological      92            876  
## 2 Natural      Climatological  92            1826  
## 3 Natural      Geophysical     9.5           253  
## 4 Natural      Hydrological    5             335  
## 5 Natural      Meteorological  3             196  
## 6 Technological Industrial accident 1             243  
## 7 Technological Miscellaneous accident 1             64  
## 8 Technological Transport        1            367
```

```
# pivoting the data to create better plot  
temp_long <- temp_df %>%  
  pivot_longer(  
    cols = c(MIN_DURATION, MEDIAN_DURATION,  
             MAX_DURATION),  
    names_to = "Duration_Type",  
    values_to = "Duration"  
  )  
  
ggplot(temp_long, aes(x = DISASTER_SUBGROUP, y = Duration,  
                     group = Duration_Type)) +  
  geom_line(aes(linetype = Duration_Type, color = Duration_Type),  
            line_width = 1) +  
  geom_point(aes(shape = Duration_Type, color = Duration_Type), size = 2) +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  labs(  
    title = "Disaster Duration Summary by Disaster Subgroup",  
    x = "Disaster Subgroup",  
    y = "Duration (Days)",  
    color = "Duration Metric",  
    linetype = "Duration Metric",  
    shape = "Duration Metric"  
  )
```



As rightly captured by the line plot, disaster subgroups “Biological” and “Climatological” have the maximum median disaster duration, and spanned the most as well. Hence, drilling down a level further to the disaster sub-type below.

```
temp_df <- df %>% filter(DURATION_FLAG == 1) %>%
  group_by(DISASTER_GROUP, DISASTER_SUBGROUP, DISASTER_TYPE) %>%
  summarize(MIN_DURATION = min(DISASTER_DURATION),
            MEDIAN_DURATION = median(DISASTER_DURATION),
            MAX_DURATION = max(DISASTER_DURATION)) %>%
  arrange(desc(MEDIAN_DURATION))
head(temp_df %>% select(DISASTER_GROUP, DISASTER_SUBGROUP,
                       DISASTER_TYPE, MEDIAN_DURATION, MAX_DURATION))
```

```
## # A tibble: 6 x 5
## # Groups:   DISASTER_GROUP, DISASTER_SUBGROUP [5]
##   DISASTER_GROUP DISASTER_SUBGROUP DISASTER_TYPE MEDIAN_DURATION MAX_DURATION
##   <chr>          <chr>          <chr>          <dbl>          <dbl>
## 1 Natural        Climatological    Drought          274            1826
## 2 Natural        Biological        Epidemic          92.5           876
## 3 Natural        Biological        Infestation       92             212
## 4 Natural        Meteorological    Extreme tempe~    22             196
## 5 Technological   Industrial accident Industrial ac~    16              30
## 6 Natural        Geophysical       Volcanic acti~    13             253
```

The table displays disaster type ‘Drought’ as the longest duration disaster on average, followed by ‘Epidemic’. Coincidentally, these two subtypes have instances as longest occurring ones. An exploration into drought and epidemic will be carried out in the later sections.

A.2 Analysis by Region

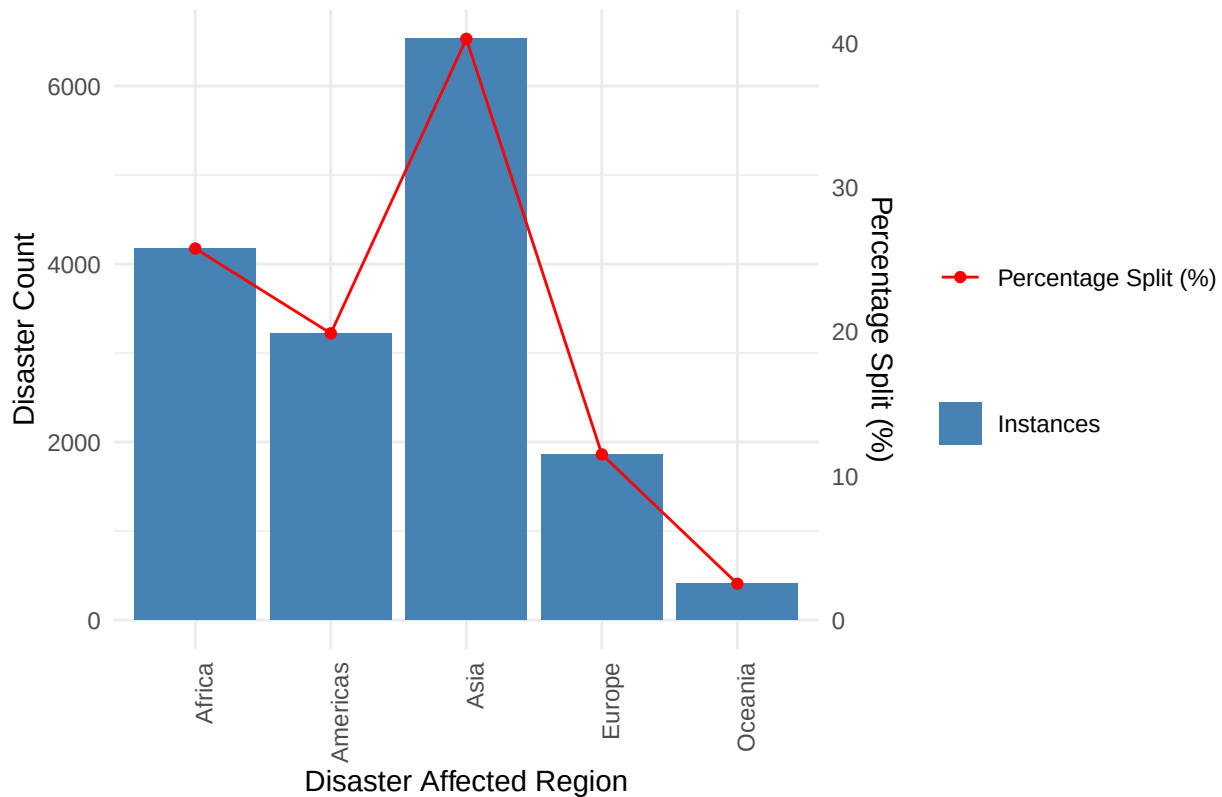
- 1) Understanding the disaster instances by region and subregion

```
temp_df <- df %>% group_by(REGION) %>%  
  summarize(INSTANCES = n(), PCT_SPLIT = n()/nrow(df)*100)  
head(temp_df, 10)
```

```
## # A tibble: 5 x 3  
##   REGION    INSTANCES PCT_SPLIT  
##   <chr>      <int>     <dbl>  
## 1 Africa      4174      25.8  
## 2 Americas    3221      19.9  
## 3 Asia        6531      40.3  
## 4 Europe      1861      11.5  
## 5 Oceania      407       2.51
```

```
ggplot(temp_df, aes(x = REGION)) +  
  geom_bar(aes(y = INSTANCES, fill = "Instances"), stat = "identity") +  
  geom_line(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)",  
                group = 1) +  
  geom_point(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)")) +  
  scale_y_continuous(  
    name = "Disaster Count",  
    sec.axis = sec_axis(~ . / scale_factor, name = "Percentage Split (%)")  
  ) +  
  scale_fill_manual(name = "", values = c("Instances" = "steelblue")) +  
  scale_color_manual(name = "", values = c("Percentage Split (%)" = "red")) +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +  
  labs(  
    title = "Instances and Percentage Split by Disaster Affected Region",  
    x = "Disaster Affected Region"  
  )
```

Instances and Percentage Split by Disaster Affected Region



Since, regions ‘Asia’, ‘Africa’ and ‘Americas’ have the highest peak in the bar chart, we’ll drill down a level to sub regions.

```
temp_df <- df %>% group_by(REGION ,SUBREGION) %>%
  summarize(INSTANCES = n()) %>%
  arrange(desc(INSTANCES))
head(temp_df,10)
```

```
## # A tibble: 10 x 3
## # Groups:   REGION [4]
##   REGION SUBREGION INSTANCES
##   <chr>   <chr>      <int>
## 1 Africa Sub-Saharan Africa 3531
## 2 Americas Latin America and the Caribbean 2400
## 3 Asia Southern Asia 2158
## 4 Asia Eastern Asia 1848
## 5 Asia South-eastern Asia 1748
## 6 Americas Northern America 821
## 7 Europe Eastern Europe 700
## 8 Africa Northern Africa 643
## 9 Asia Western Asia 633
## 10 Europe Southern Europe 617
```

This drill down version of analysis throw light upon subregions ‘Sub-Saharan Africa’, ‘Latin America and the Carribean’, ‘Southern Asia’ and ‘Eastern Asia’ as the most impacted sub regions. A detailed analyses on these subregions will follow in the later sections.

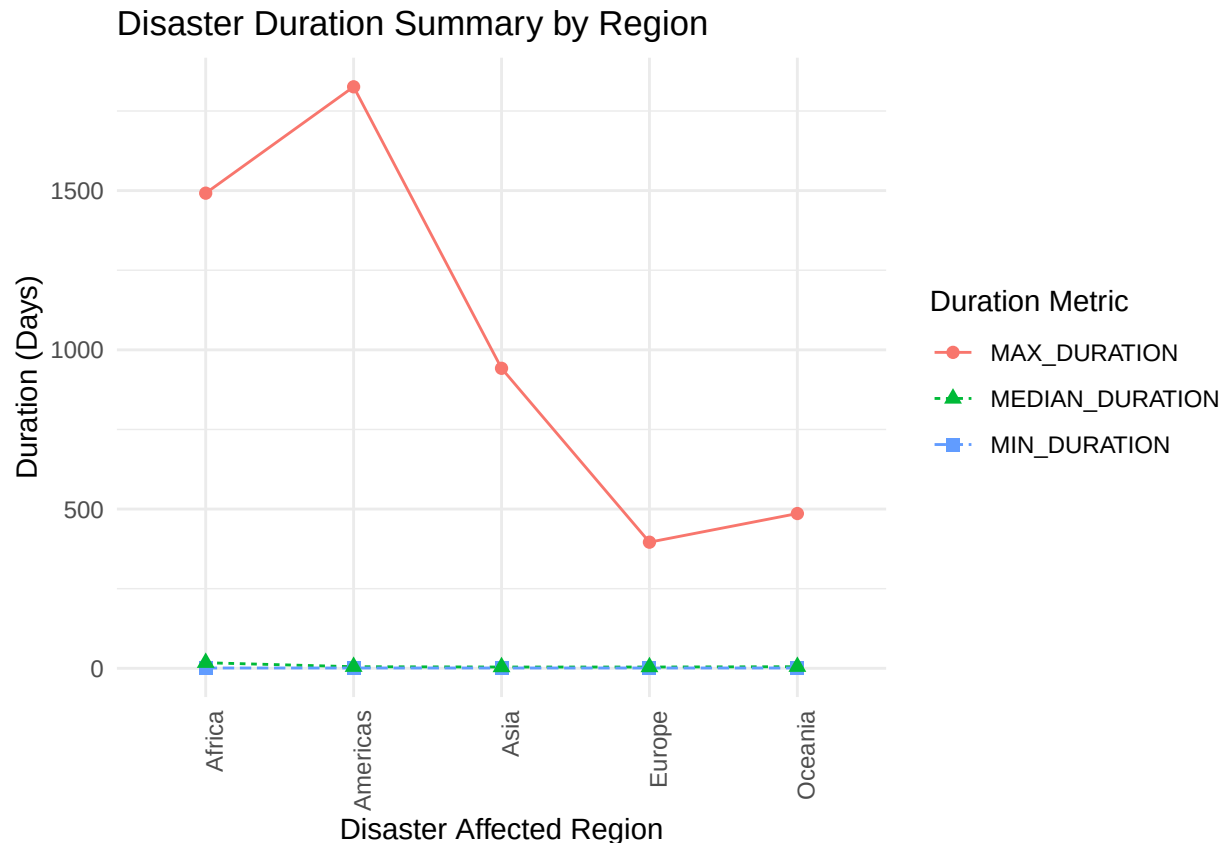
2) Understanding the disaster duration by region and subregion

```
temp_df <- df %>% filter(DURATION_FLAG == 1) %>%
  group_by(REGION) %>%
  summarize(MIN_DURATION = min(DISASTER_DURATION),
            MEDIAN_DURATION = median(DISASTER_DURATION),
            MAX_DURATION = max(DISASTER_DURATION))
# considering only valid cases for disaster duration cases
head(temp_df %>%
  select(c(REGION, MEDIAN_DURATION, MAX_DURATION)), 10)
```

```
## # A tibble: 5 x 3
##   REGION    MEDIAN_DURATION MAX_DURATION
##   <chr>          <dbl>         <dbl>
## 1 Africa          17.5          1492
## 2 Americas         5          1826
## 3 Asia            4           942
## 4 Europe          4           396
## 5 Oceania         5           486
```

```
# pivoting the data to create better plot
temp_long <- temp_df %>%
  pivot_longer(
    cols = c(MIN_DURATION, MEDIAN_DURATION,
              MAX_DURATION),
    names_to = "Duration_Type",
    values_to = "Duration"
  )

ggplot(temp_long, aes(x = REGION, y = Duration,
                      group = Duration_Type)) +
  geom_line(aes(linetype = Duration_Type, color = Duration_Type),
            line_width = 1) +
  geom_point(aes(shape = Duration_Type, color = Duration_Type), size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Disaster Duration Summary by Region",
    x = "Disaster Affected Region",
    y = "Duration (Days)",
    color = "Duration Metric",
    linetype = "Duration Metric",
    shape = "Duration Metric"
  )
```

Analysis of regions by duration again highlight region 'Africa' as the region having been impacted by disasters the longest on average.

A.3 Analysis by Year

1) Understanding the disaster instances by disaster start year

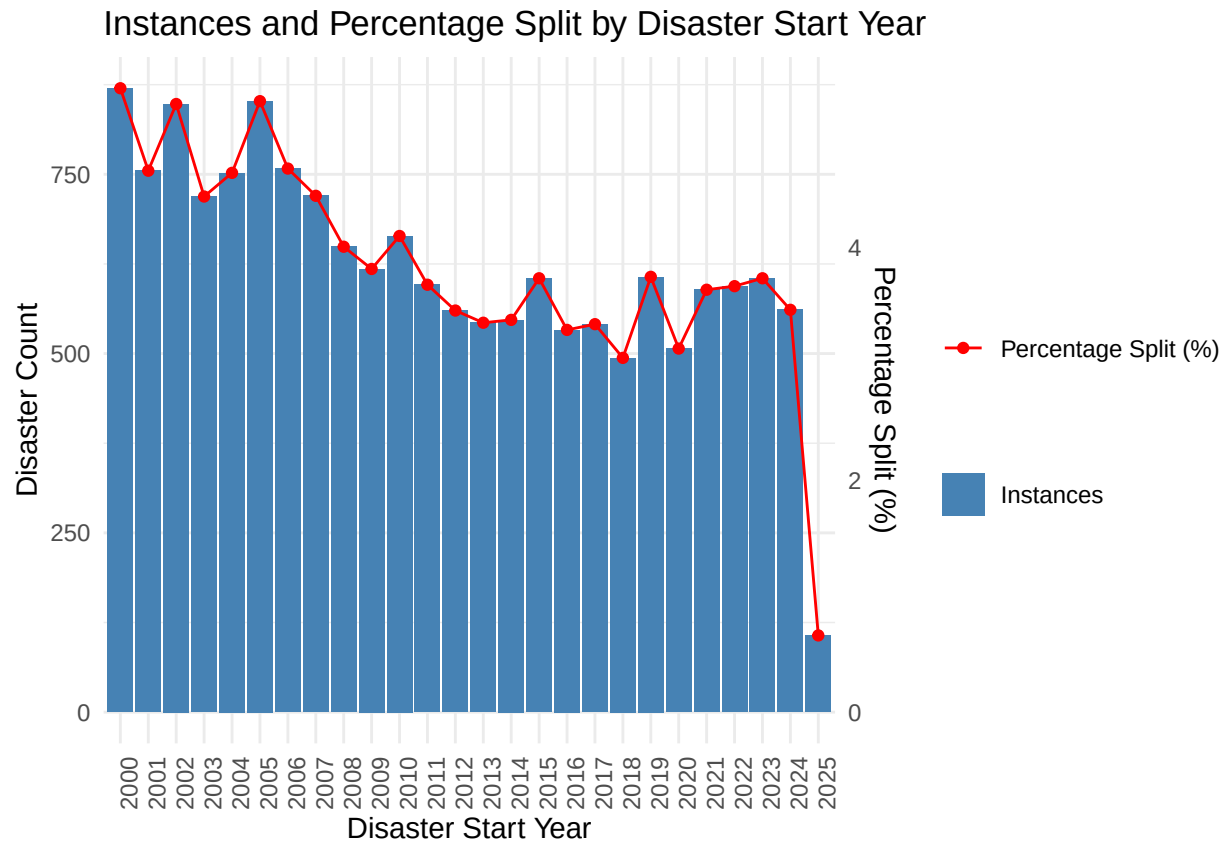
```
temp_df <- df %>% group_by(START_YEAR) %>%
  summarize(INSTANCES = n(), PCT_SPLIT = n()/nrow(df)*100)

ggplot(temp_df, aes(x = factor(START_YEAR))) +
  geom_bar(aes(y = INSTANCES, fill = "Instances"), stat = "identity") +
  geom_line(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)",
    group = 1) +
  geom_point(aes(y = PCT_SPLIT * scale_factor, color = "Percentage Split (%)")) +
  scale_y_continuous(
    name = "Disaster Count",
    sec.axis = sec_axis(~ . / scale_factor, name = "Percentage Split (%)")
  ) +
  scale_fill_manual(name = "", values = c("Instances" = "steelblue")) +
  scale_color_manual(name = "", values = c("Percentage Split (%)" = "red")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
```

```

title = "Instances and Percentage Split by Disaster Start Year",
x = "Disaster Start Year"
)

```



The distribution of count of disasters was high between 2000 to 2005. The period between 2006 to 2010 saw a steady decline in the instances of disasters. Post 2010, the distribution looks more or less uniform. An interesting avenue to explore could be viewing disasters in 3 periods: 2000-2005, 2006-2010 and Post 2010.

B. Two-Dimensional Summaries

These two-dimensional summaries explore relationships between disaster characteristics such as classification, region, and start year to uncover patterns and trends across the datasets.

B.1 Analysis by Disaster Classification and Region

1) Understanding disasters having maximum median duration : Drought and Epidemic

```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Drought','Epidemic')) %>%  
  group_by(DISASTER_TYPE,REGION) %>% summarise(INSTANCES = n())  
temp_df <- temp_df %>% pivot_wider(values_from='INSTANCES',names_from='REGION')  
head(temp_df)
```

```
## # A tibble: 2 x 6  
## # Groups:   DISASTER_TYPE [2]  
##   DISASTER_TYPE Africa Americas Asia Europe Oceania  
##   <chr>          <int>    <int> <int>  <int>  <int>  
## 1 Drought        176      102   94    26    18  
## 2 Epidemic       598      82   159   17    19
```

With a further drill down, in the region ‘African’, the sub region ‘Sub-Saharan Africa’ is the most affected with 172 instances of drought out of 176 total, and 518 epidemic instances (349 bacterial and 169 viral instances) out of a total of 598 epidemic instances in the African region.

```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Drought','Epidemic')) %>%  
  group_by(DISASTER_TYPE,REGION) %>% summarise(TOTAL_AFFECTED =  
                                                sum(NO__AFFECTED,na.rm = TRUE))  
temp_df <- temp_df %>% pivot_wider(values_from='TOTAL_AFFECTED',  
                                   names_from='REGION')  
head(temp_df)
```

```
## # A tibble: 2 x 6  
## # Groups:   DISASTER_TYPE [2]  
##   DISASTER_TYPE Africa Americas Asia Europe Oceania  
##   <chr>          <int>    <int>  <int>  <int>  <int>  
## 1 Drought    450675157 59439080 1178556850 1304769 2870192  
## 2 Epidemic    5138054 2765245 1407660 17142 35064
```

Zeroing in on sub region ‘Sub-Saharan Africa’, 432.8 million people were affected due to droughts and 5.9 million affected by epidemics.

Another notable combination is that of droughts in Asia, which affected 1.17 billion people. Drilling down at sub region level, it is observed that in sub region ‘Eastern Asia’ 359 million and in sub region ‘Southern Asia’ 735 million people were affected due to droughts.

Two major drought incidents of drought occurred in country ‘India’ under the sub region ‘Southern Asia’:

- Disaster ID: 2015-9618-IND, a drought which started in 2015, went on for more than 2 years and affected 330 million people.
- Disaster ID: 2002-9349-IND, a drought which started in July 2002, affected 330 million people. This instances was also declared as a national emergency.

For more information, kindly visit: [Droughts in India](#) and [Droughts in India](#)

One anomaly is that region ‘Americas’, which is generally seen as a developed portion of the world also experienced major drought episodes with 59 million people in sub region ‘Latin America and the Caribbean’ getting affected.

```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Drought','Epidemic')) %>%
  group_by(DISASTER_TYPE,REGION) %>% summarise(TOTAL_DEATHS =
                                              sum(TOTAL_DEATHS,na.rm = TRUE))
temp_df <- temp_df %>% pivot_wider(values_from='TOTAL_DEATHS',
                                   names_from='REGION')
head(temp_df)
```

```
## # A tibble: 2 x 6
## # Groups:   DISASTER_TYPE [2]
##   DISASTER_TYPE Africa Americas Asia Europe Oceania
##   <chr>          <int>    <int> <int>  <int>  <int>
## 1 Drought        23374      481   279     2     24
## 2 Epidemic        97683     10270 14034    18     465
```

Though much of the data wrt Deaths is missing in region ‘Africa’, it seems, with whatever data is available, that most people died again in the ‘Sub-Saharan Africa’ sub region with 37 k deaths due to viral epidemics and 52 k deaths due to bacterial epidemics.

```
temp_df <- df %>% filter((DISASTER_TYPE %in% c('Drought','Epidemic')) &
                        (OFDA_BHA_RESPONSE=='Yes')) %>%
  group_by(DISASTER_TYPE,REGION) %>% summarise(INSTANCES_RESPONDED = n())
temp_df <- temp_df %>% pivot_wider(values_from='INSTANCES_RESPONDED',
                                   names_from='REGION')
head(temp_df)
```

```
## # A tibble: 2 x 6
## # Groups:   DISASTER_TYPE [2]
##   DISASTER_TYPE Africa Americas Asia Europe Oceania
##   <chr>          <int>    <int> <int>  <int>  <int>
## 1 Drought         43      2     8     1     4
## 2 Epidemic        26      3    NA    NA     1
```

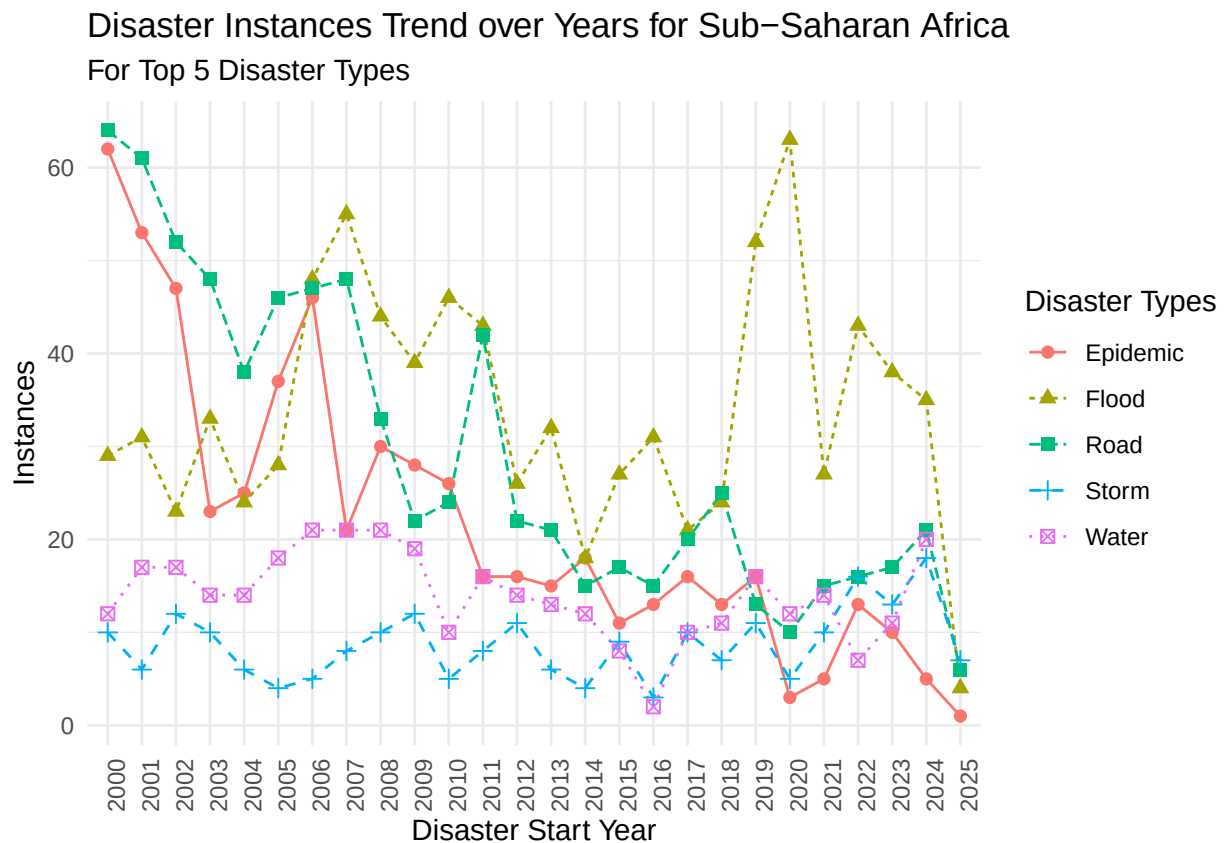
As expected, the sub regions ‘Sub-Saharan Africa’ and ‘Latin America and the Caribbean’ received the most responses from Office of US Foreign Disaster Assistance (OFDA) or the Bureau of Humanitarian Assistance (BHA).

Region Deep Dive: Sub-Saharan Africa

Analysis of Disasters having the maximum median duration directs one to sub region ‘Sub-Saharan Africa’, a region below the equator and a largely underdeveloped region in the world, as the most impacted sub-region with a record count of 3541 disasters with 884 floods, 758 road accidents, 569 epidemics, 350 road accidents, 226 storms and 172 droughts.

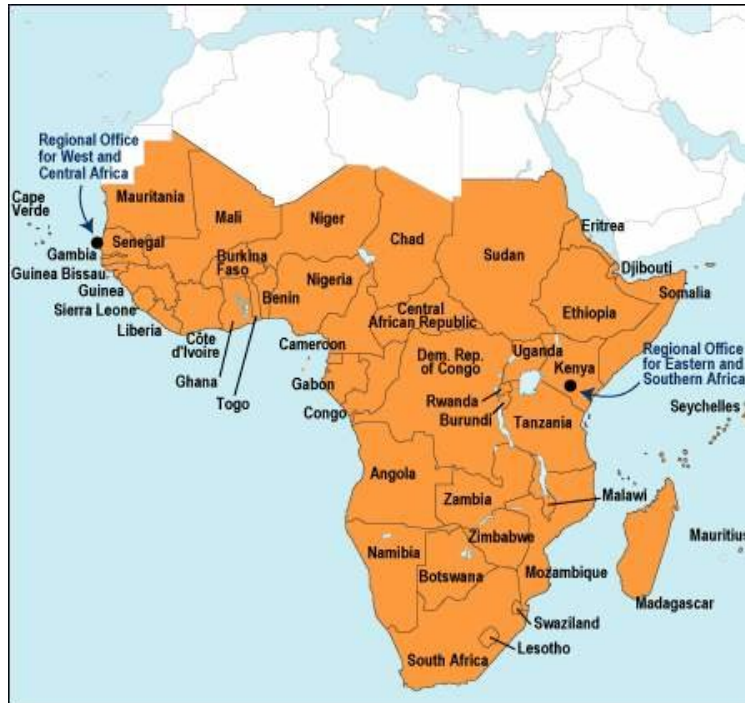
```
temp_df <- df %>% filter(SUBREGION %in% c('Sub-Saharan Africa')) %>%
  filter(DISASTER_TYPE %in% c('Flood','Road','Epidemic','Water','Storm')) %>%
  group_by(START_YEAR, SUBREGION, DISASTER_TYPE) %>% summarize(INSTANCES=n())
```

```
ggplot(temp_df, aes(x = factor(START_YEAR), y = INSTANCES,
                    group = DISASTER_TYPE)) +
  geom_line(aes(linetype = DISASTER_TYPE, color = DISASTER_TYPE),
            line_width = 1) +
  geom_point(aes(shape = DISASTER_TYPE, color = DISASTER_TYPE), size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Disaster Instances Trend over Years for Sub-Saharan Africa",
    subtitle = "For Top 5 Disaster Types",
    x = "Disaster Start Year",
    y = "Instances",
    color = "Disaster Types",
    linetype = "Disaster Types",
    shape = "Disaster Types"
  )
)
```

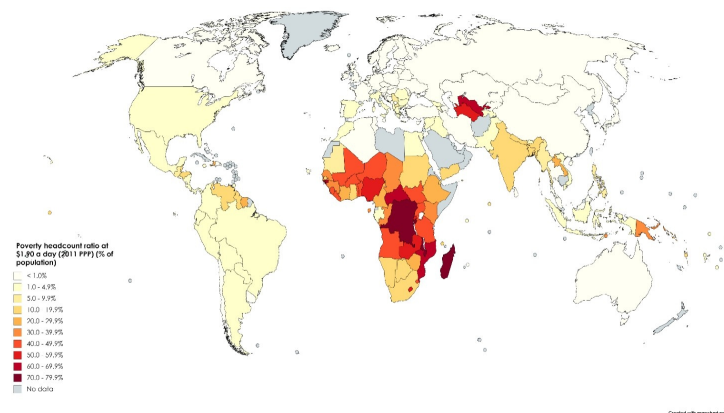


From the trend visual, following summaries can be inferred:

- The sub region is constantly affected by floods over the years with a peak in 2020
- Sub-Saharan Africa seems to have done better over the years to reduce the epidemic instances. Similar inference can be attached to the road accidents with the help of improved infrastructure.
- Though the numbers for Storms and Water based accidents seem to be low, the region is constantly affected by these over the years.



Map of Sub-Saharan Africa



Poverty Headcount Ratio Map

The poverty map shows the poverty ratio for the 'Sub-Saharan Africa' sub region ranging between 30% to 80%!!

2) Understanding disaster having maximum instances : Flood

```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Flood')) %>%
  group_by(DISASTER_TYPE, REGION) %>% summarise(INSTANCES = n())
temp_df <- temp_df %>% pivot_wider(values_from='INSTANCES', names_from='REGION')
head(temp_df)
```

```
## # A tibble: 1 x 6
## # Groups:   DISASTER_TYPE [1]
##   DISASTER_TYPE Africa Americas Asia Europe Oceania
##   <chr>          <int>    <int> <int> <int>    <int>
## 1 Flood           981      900  1673   483     103
```

With a further drill down, in the region ‘Asia’, the sub regions ‘South Eastern Asia’ and ‘Southern Asia’ are the most affected with 604 and 172 instances of flood respectively. Under ‘Southern Asia’, country ‘India’ has been hit with 199 floods, and 108 of these are Riverine floods.

```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Flood')) %>%
  group_by(DISASTER_TYPE, REGION) %>% summarise(TOTAL_AFFECTED =
                                              sum(TOTAL_AFFECTED, na.rm = TRUE))
temp_df <- temp_df %>% pivot_wider(values_from='TOTAL_AFFECTED',
                                   names_from='REGION')
head(temp_df)
```

```
## # A tibble: 1 x 6
## # Groups:   DISASTER_TYPE [1]
##   DISASTER_TYPE Africa Americas Asia Europe Oceania
##   <chr>          <int>    <int> <int> <int>    <int>
## 1 Flood           95835289 72770832 1675482929 8988594 1125144
```

600 million people are affected due to floods in sub region ‘Southern Asia’ and contributing to this is country ‘India’ with 183 million people affected due to floods, appearing 7 times in the top 10 most flood affected countries.

Region Deep Dive: China

950 million people are affected in sub region ‘Eastern Asia’ with 305 flood instances. China contributes the most to it with 221 flood disasters causing distress to 943 million people. China recorded 3 riverine floods caused by torrential rains with IDs (2003-0315-CHN along the Yangtze river basin, 2010-0239-CHN, 2007-0268-CHN).

```
temp_df <- df %>% filter(COUNTRY %in% c('China')) %>%
  filter(DISASTER_TYPE %in% c('Flood', 'Earthquake', 'Explosion', 'Water', 'Storm')) %>%
  group_by(START_YEAR, COUNTRY, DISASTER_TYPE) %>% summarize(INSTANCES=n())

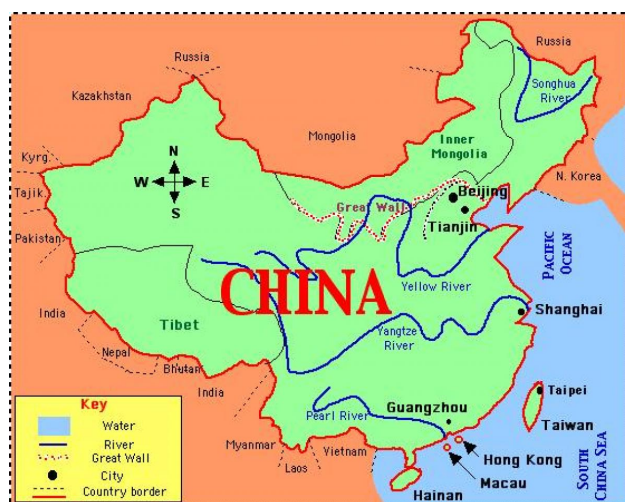
ggplot(temp_df, aes(x = factor(START_YEAR), y = INSTANCES,
                        group = DISASTER_TYPE)) +
  geom_line(aes(linetype = DISASTER_TYPE, color = DISASTER_TYPE),
            line_width = 1) +
  geom_point(aes(shape = DISASTER_TYPE, color = DISASTER_TYPE), size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Disaster Instances Trend over Years for China",
    subtitle="For Top 4 Disaster Types",
    x = "Disaster Start Year",
    y = "Instances",
    color = "Disaster Types",
    linetype = "Disaster Types",
```

```
shape = "Disaster Types"
)
```

Disaster Instances Trend over Years for China
For Top 4 Disaster Types



China seems to be most affected by floods over the years followed by storms. In 2006, there is a sharp peak in the flood instances with a figure of 20 which caused 430 deaths and affected more than 14 million people. All of the floods were caused by torrential monsoonal rains.



China Map with Rivers


```
temp_df <- df %>% filter(DISASTER_TYPE %in% c('Flood')) %>%
  group_by(DISASTER_TYPE, REGION) %>% summarise(TOTAL_DEATHS =
    sum(TOTAL_DEATHS, na.rm = TRUE))
temp_df <- temp_df %>% pivot_wider(values_from='TOTAL_DEATHS',
  names_from='REGION')
head(temp_df)
```

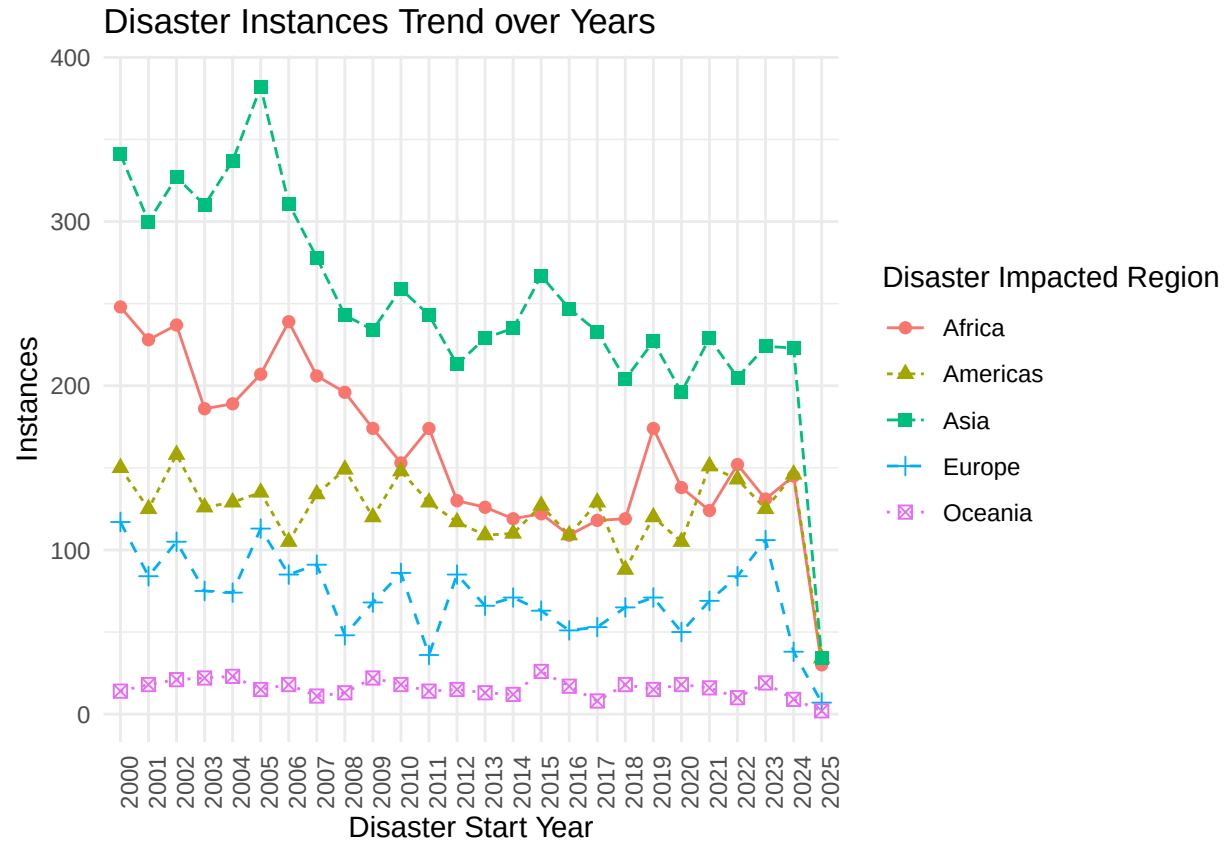
```
## # A tibble: 1 x 6
## # Groups:   DISASTER_TYPE [1]
##   DISASTER_TYPE Africa Americas Asia Europe Oceania
##   <chr>          <int>    <int> <int> <int>    <int>
## 1 Flood          25877    15468 92457  2466     273
```

Sub region 'Southern Asia' recorded the most deaths with a figure of 59.6 k deaths with India contributing 35 k deaths to this figure.

B.2 Analysis by Start Year and Region

```
temp_df <- df %>% group_by(START_YEAR, REGION) %>% summarize(INSTANCES=n())

ggplot(temp_df, aes(x = factor(START_YEAR), y = INSTANCES,
  group = REGION)) +
  geom_line(aes(linetype = REGION, color = REGION),
    line_width = 1) +
  geom_point(aes(shape = REGION, color = REGION), size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Disaster Instances Trend over Years",
    x = "Disaster Start Year",
    y = "Instances",
    color = "Disaster Impacted Region",
    linetype = "Disaster Impacted Region",
    shape = "Disaster Impacted Region"
  )
```

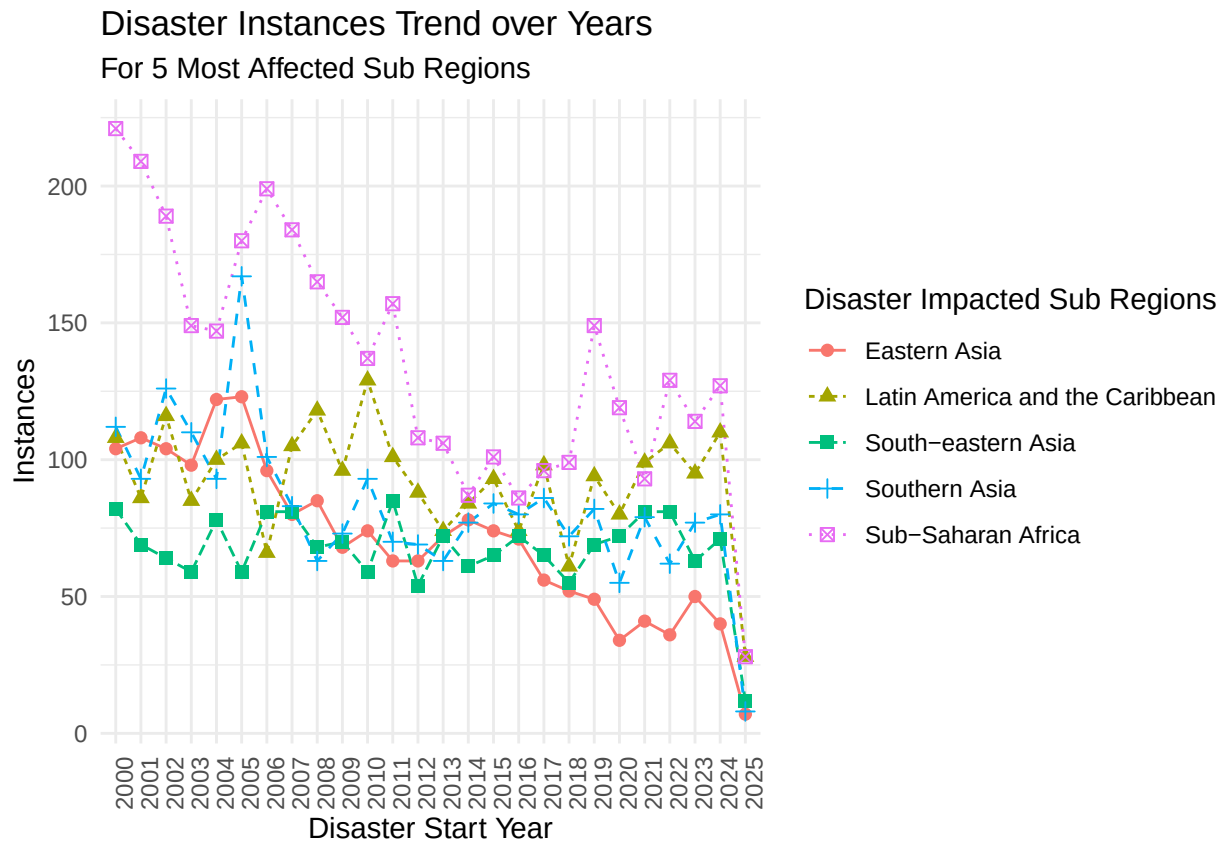


Similar to the inferences made in the above sections, this visual seconds the reason that region ‘Asia’ recorded the most disasters over the year, followed by ‘Africa’ and ‘Americas’. Regions ‘Oceania’ and ‘Europe’ seem to be safe from disasters. This could be by their readiness to the disasters, or the geographical location.

Drilling down a level further at the sub region level and observing the trends of top 5 worst affected sub regions.

```
temp_df <- df %>% filter(SUBREGION %in% c('Sub-Saharan Africa',
                                           'Latin America and the Caribbean',
                                           'Eastern Asia','South-eastern Asia',
                                           'Southern Asia')) %>%
  group_by(START_YEAR, SUBREGION) %>% summarize(INSTANCES=n())
ggplot(temp_df, aes(x = factor(START_YEAR), y = INSTANCES,
                      group = SUBREGION)) +
  geom_line(aes(linetype = SUBREGION, color = SUBREGION),
            line_width = 1) +
  geom_point(aes(shape = SUBREGION, color = SUBREGION), size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Disaster Instances Trend over Years",
    subtitle = 'For 5 Most Affected Sub Regions',
    x = "Disaster Start Year",
    y = "Instances",
    color = "Disaster Impacted Sub Regions",
    linetype = "Disaster Impacted Sub Regions",
```

```
shape = "Disaster Impacted Sub Regions"
)
```



Clearly, 'Sub Saharan Africa' remained the most affected region over the years, followed by 'Latin America and the Caribbean'. Asiatic subregions recorded higher instances of disasters in the early 2000's, but later seem to have readied themselves against disasters.

C. TopN Deep-Dives

This section presents focused deep-dive analyses into the top contributors to total deaths, examined by disaster classification, region, and start year to identify the most impactful events and trends.

C.1 Analysis of Total Deaths by Disaster Classification

```
total_deaths = sum(df$TOTAL_DEATHS, na.rm=TRUE)

temp_df <- df %>% group_by(DISASTER_GROUP, DISASTER_SUBGROUP, DISASTER_TYPE) %>%
  summarize(TOTAL_DEATHS = sum(TOTAL_DEATHS, na.rm=TRUE),
            PCT_DEATHS = sum(TOTAL_DEATHS, na.rm=TRUE)/total_deaths*100,
            .groups='drop') %>%
  slice_max(n=5, order_by=TOTAL_DEATHS)
temp_df
```

```
## # A tibble: 5 x 5
##   DISASTER_GROUP DISASTER_SUBGROUP DISASTER_TYPE      TOTAL_DEATHS PCT_DEATHS
##   <chr>          <chr>          <chr>          <int>        <dbl>
## 1 Natural      Geophysical      Earthquake      794514        44.2
## 2 Natural      Meteorological   Extreme temperature 289981        16.1
## 3 Natural      Meteorological   Storm           223227        12.4
## 4 Natural      Hydrological     Flood           136541         7.60
## 5 Natural      Biological       Epidemic        122470         6.81
```

Out of the total death records available, 87.15% of the total deaths are recorded by Top 5 Disaster Sub Types.

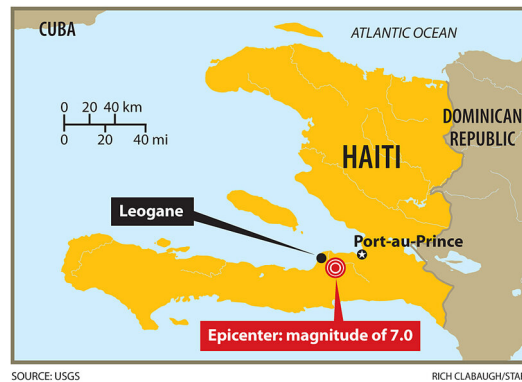
Disaster Type Deep Dive: Earthquake

Earthquake alone accounts for 44.20% of deaths. Deep diving into the details of most disastrous earthquakes as follows:

```
temp_df <- df %>% filter(DISASTER_TYPE=='Earthquake') %>% arrange(desc(TOTAL_DEATHS)) %>%
  select(SUBREGION, COUNTRY, MAGNITUDE, MAGNITUDE_SCALE, TOTAL_DEATHS) %>%
  slice_max(order_by=TOTAL_DEATHS, n=5)
temp_df
```

```
##           SUBREGION  COUNTRY MAGNITUDE  MAGNITUDE_SCALE
## 1 Latin America and the Caribbean    Haiti      7.0 Moment Magnitude
## 2           South-eastern Asia Indonesia      9.1 Moment Magnitude
## 3           Eastern Asia      China      7.9 Moment Magnitude
## 4           Southern Asia  Pakistan      7.6 Moment Magnitude
## 5           Western Asia    Türkiye      7.8 Moment Magnitude
##   TOTAL_DEATHS
## 1      222570
## 2      165708
## 3       87476
## 4       73338
## 5       53000
```

- Haiti 2010 Earthquake : A devastating earthquake (disaster ID: 2010-0017-HTI) of magnitude 7.0 occurred in Haiti (sub region 'Latin America and the Caribbean') on 12th January 2010 leading to death of 222,750 people and affecting more than 3.7 million people.



2010 Earthquake Epicenter in Haiti

To read more about the disaster, please visit: [Haiti 2010 Earthquake](#)

- Indian Ocean 2004 Earthquake : Followed up separately as a use case.

C.2 Analysis of Total Deaths by Disaster Start Year

```
total_deaths = sum(df$TOTAL_DEATHS, na.rm=TRUE)

temp_df <- df %>% group_by(START_YEAR) %>%
  summarize(TOTAL_DEATHS = sum(TOTAL_DEATHS, na.rm=TRUE)/10000,
            PCT_DEATHS = sum(TOTAL_DEATHS*10000, na.rm=TRUE)/total_deaths*100,
            .groups='drop')
temp_df %>% slice_max(order_by = TOTAL_DEATHS, n=3)
```

```
## # A tibble: 3 x 3
##   START_YEAR TOTAL_DEATHS PCT_DEATHS
##   <int>      <dbl>      <dbl>
## 1    2010      33.7      18.7
## 2    2004      25.5      14.2
## 3    2008      24.9      13.8
```

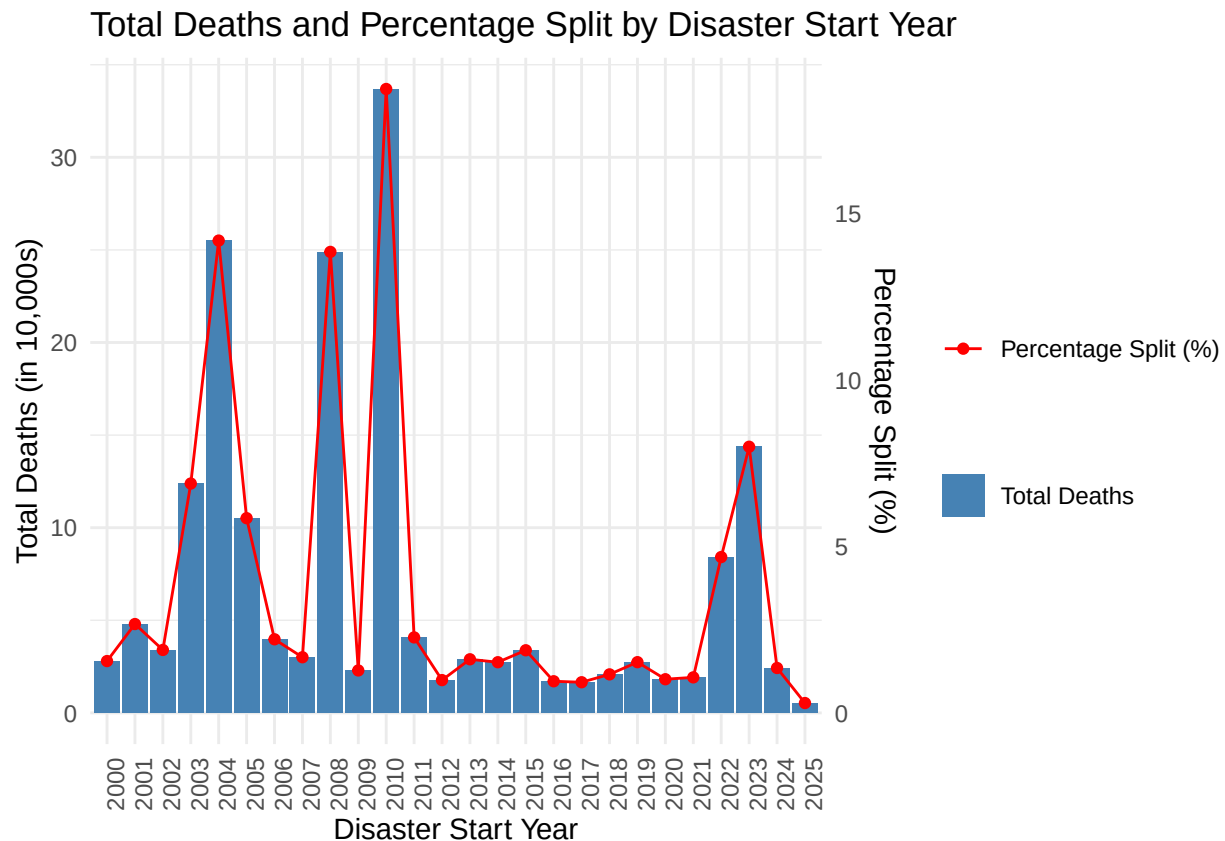
The biggest contributions to the top 3 years are made by the following disasters:

- Haiti Earthquake in 2010
- Indian Ocean Earthquake Tsunami deaths in Indonesia in 2004
- Tropical cyclone with wind speed of 215 kmph in Myanmar in 2008

```

scale_factor <- max(temp_df$TOTAL_DEATHS, na.rm = TRUE) / max(temp_df$PCT_DEATHS, na.rm = TRUE)
ggplot(temp_df, aes(x = factor(START_YEAR))) +
  geom_bar(aes(y = TOTAL_DEATHS, fill = "Total Deaths"), stat = "identity") +
  geom_line(aes(y = PCT_DEATHS * scale_factor, color = "Percentage Split (%)",
                group = 1) +
  geom_point(aes(y = PCT_DEATHS * scale_factor, color = "Percentage Split (%)")) +
  scale_y_continuous(
    name = "Total Deaths (in 10,000s)",
    sec.axis = sec_axis(~ . / scale_factor, name = "Percentage Split (%)")
  ) +
  scale_fill_manual(name = "", values = c("Total Deaths" = "steelblue")) +
  scale_color_manual(name = "", values = c("Percentage Split (%)" = "red")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(
    title = "Total Deaths and Percentage Split by Disaster Start Year",
    x = "Disaster Start Year"
  )

```



With 17,694 deaths in 2012 as compared to other years certainly proves the **End of the World in 2012** claim wrong! Still, this figure of 17.6 K deaths in 2012 deaths is not something to be cheered

D. Case Studies

This section presents detailed case studies, including the 2004 Indian Ocean tsunami, an in-depth disaster analysis of India.

D.1 Disasters in India

India has been severely impacted by three major disasters: floods due to its vast network of rivers, droughts due to dependency on monsoons and earthquake. An overall view by disaster type is given below:

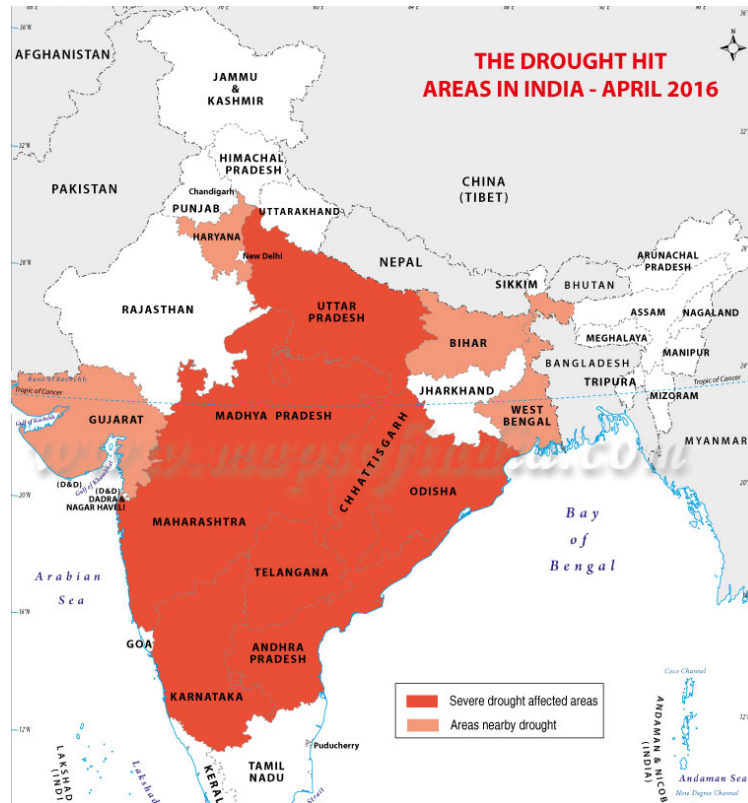
```
temp_df <- df %>% filter(COUNTRY=='India') %>%
  group_by(DISASTER_GROUP, DISASTER_SUBGROUP, DISASTER_TYPE) %>%
  summarise(INSTANCES = n(), TOTAL_DEATHS = sum(TOTAL_DEATHS, na.rm=TRUE),
            TOTAL_AFFECTED = sum(TOTAL_AFFECTED, na.rm=TRUE)) %>%
  arrange(desc(TOTAL_DEATHS))
head(temp_df, 10)
```

```
## # A tibble: 10 x 6
## # Groups:   DISASTER_GROUP, DISASTER_SUBGROUP [6]
##   DISASTER_GROUP DISASTER_SUBGROUP DISASTER_TYPE INSTANCES TOTAL_DEATHS
##   <chr>          <chr>          <chr>          <int>    <int>
## 1 Natural       Geophysical       Earthquake      11      37928
## 2 Natural       Hydrological      Flood           199     35270
## 3 Natural       Meteorological    Extreme tempera~ 35     10398
## 4 Natural       Meteorological    Storm           103     4315
## 5 Technological Transport         Road            143     4065
## 6 Technological Transport         Rail             56     1940
## 7 Natural       Biological        Epidemic        25     1693
## 8 Natural       Hydrological      Mass movement (~ 31     1637
## 9 Technological Miscellaneous accident Miscellaneous a~ 33     1627
## 10 Technological Transport         Water            40     1501
## # i 1 more variable: TOTAL_AFFECTED <int>
```

Droughts

Two major drought incidents of drought occurred in country 'India':

- Disaster ID: 2015-9618-IND, a drought which started in 2015, went on for more than 2 years and affected 330 million people.
- Disaster ID: 2002-9349-IND, a drought which started in July 2002, affected 330 million people. This instances was also declared as a national emergency.
(Since, total_deaths record is not available for droughts in India, these two are the most devastating wrt the number of affected people.)



Drought Affected Areas of India in 2016 (Reference: Maps of India)

The map shows the entire Indian mainland severely affected by droughts and other regions indirectly affected. Kindly note that Indian mainland is the grain producing land for India.

Floods

Three major incidents of flood in India by number of deaths are:

- Disaster ID: 2002-0374-IND causing 549 deaths
- Disaster ID: 2004-0298-IND causing 900 deaths
- Disaster ID: 2000-0610-IND causing 884 deaths All of these 3 floods occurred in the Ganges river basin due to torrential monsoonal rains.



Major Rivers of India (Reference: Maps of India)

Earthquake

Two major incidents of earthquakes in India by number of deaths are:

- Disaster ID: 2001-0033-IND, a 7.7 magnitude earthquake causing more than 20k deaths. Read more at 2001 Gujarat Earthquake
- Disaster ID: 2004-0659-IND, a 9.1 magnitude earthquake causing more than 16.3k deaths. Read more at 2004 Indian Ocean Earthquake

Disasters in India over Years

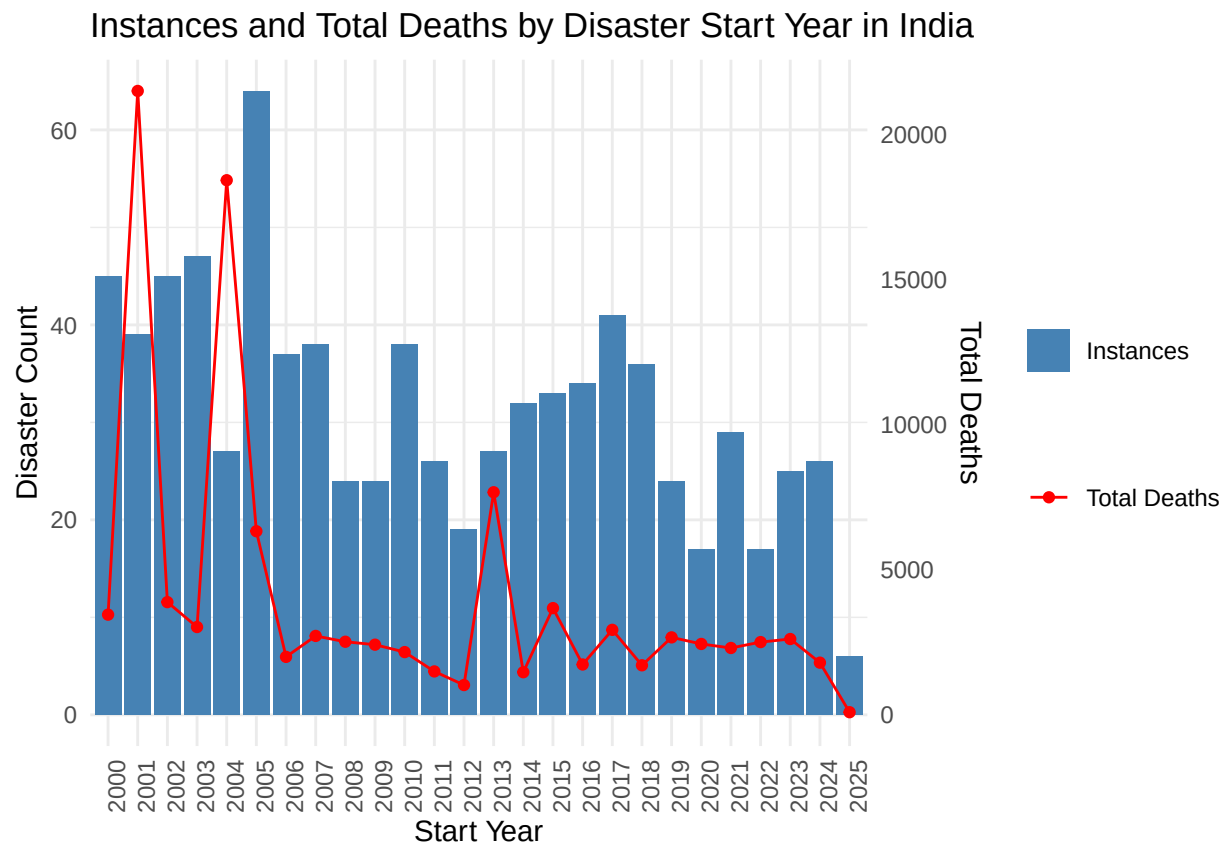
```
temp_df <- df %>% filter(COUNTRY=='India') %>%
  group_by(START_YEAR) %>%
  summarise(INSTANCES = n(), TOTAL_DEATHS = sum(TOTAL_DEATHS, na.rm=TRUE),
            TOTAL_AFFECTED = sum(TOTAL_AFFECTED, na.rm=TRUE)) %>%
  arrange(desc(TOTAL_DEATHS))
scale_factor <- max(temp_df$INSTANCES, na.rm = TRUE) / max(temp_df$TOTAL_DEATHS, na.rm = TRUE)

ggplot(temp_df, aes(x = factor(START_YEAR))) +
  geom_bar(aes(y = INSTANCES, fill = "Instances"), stat = "identity") +
  geom_line(aes(y = TOTAL_DEATHS * scale_factor, color = "Total Deaths"), group = 1) +
  geom_point(aes(y = TOTAL_DEATHS * scale_factor, color = "Total Deaths")) +
```

```

scale_y_continuous(
  name = "Disaster Count",
  sec.axis = sec_axis(~ . / scale_factor, name = "Total Deaths")
) +
scale_fill_manual(name = "", values = c("Instances" = "steelblue")) +
scale_color_manual(name = "", values = c("Total Deaths" = "red")) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
labs(
  title = "Instances and Total Deaths by Disaster Start Year in India",
  x = "Start Year")

```



Clearly, years 2001 and 2004 show a high spike in the total number of deaths and their corresponding disasters have been discussed above already. As far as the number of disasters is concerned, in year 2005, the peak is created by 17 instances of floods and 12 episodes of road accidents.

D.2 The 2004 Indian Ocean Earthquake and Tsunami

```

temp_df <- df %>% filter(DISASTER_TYPE=='Earthquake',
                        MAGNITUDE_SCALE=='Moment Magnitude',
                        MAGNITUDE==9.1, START_DATE=='2004-12-26') %>%
  arrange(desc(TOTAL_DEATHS))

```

```
temp_df %>% select(DISASTER_SUBTYPE,SUBREGION,COUNTRY, TOTAL_AFFECTED,
TOTAL_DEATHS,)
```

```
##   DISASTER_SUBTYPE      SUBREGION      COUNTRY
## 1      Tsunami South-eastern Asia      Indonesia
## 2      Tsunami      Southern Asia      Sri Lanka
## 3      Tsunami      Southern Asia      India
## 4      Tsunami South-eastern Asia      Thailand
## 5      Tsunami Sub-Saharan Africa      Somalia
## 6      Tsunami      Southern Asia      Maldives
## 7      Tsunami South-eastern Asia      Malaysia
## 8      Tsunami South-eastern Asia      Myanmar
## 9      Tsunami Sub-Saharan Africa United Republic of Tanzania
## 10     Tsunami Sub-Saharan Africa      Seychelles
## 11     Tsunami      Southern Asia      Bangladesh
## 12     Tsunami Sub-Saharan Africa      Kenya
##   TOTAL_AFFECTED TOTAL_DEATHS
## 1           532898       165708
## 2          1019306       35399
## 3           654512       16389
## 4            67007       8345
## 5          105083        298
## 6           27214        102
## 7            5063         80
## 8           15700         71
## 9              NA         10
## 10           4830          3
## 11              NA          2
## 12              NA          1
```

The 2004 Indian Ocean Tsunami, triggered by a magnitude 9.1 earthquake off the coast of Sumatra, resulted in over 226,408 deaths and affected over 2.4 million people across 12 countries, making it the deadliest tsunami in recorded history. The earthquake, which occurred on December 26, 2004, caused significant damage to coastal regions in countries like Indonesia, Thailand, Sri Lanka, and India.

In terms of loss of life, it wreaked havoc in island nations of 'Indonesia' and 'Sri Lanka', hitting the latter from both fronts western and eastern fronts due to ripples effect.

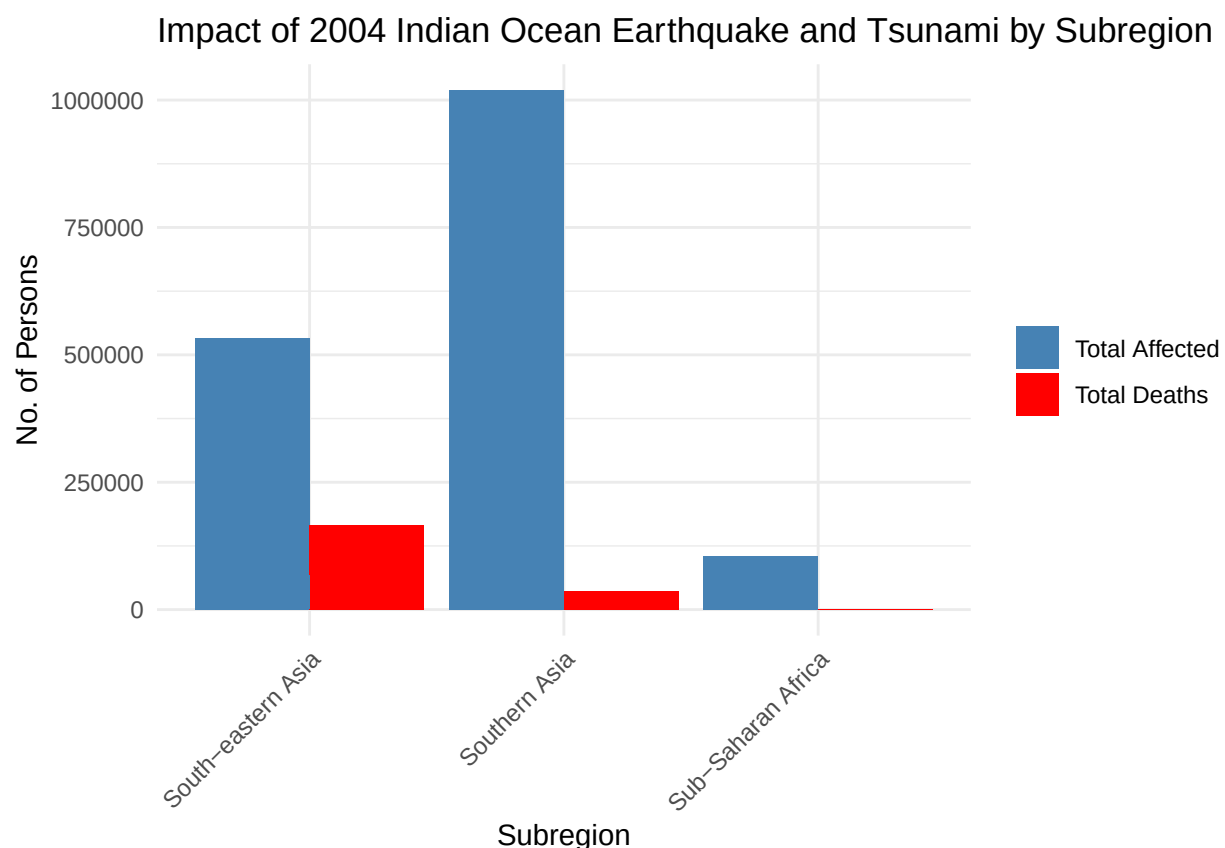
```
temp_df %>% group_by(SUBREGION) %>% summarise(TOTAL_AFFECTED=sum(TOTAL_AFFECTED, na.rm=TRUE),
TOTAL_DEATHS= sum(TOTAL_DEATHS, na.rm=TRUE))
```

```
## # A tibble: 3 x 3
##   SUBREGION      TOTAL_AFFECTED TOTAL_DEATHS
##   <chr>          <int>         <int>
## 1 South-eastern Asia      620668      174204
## 2 Southern Asia          1701032      51892
## 3 Sub-Saharan Africa      109913        312
```

```
temp_long <- temp_df %>%
  pivot_longer(cols = c(TOTAL_AFFECTED, TOTAL_DEATHS),
    names_to = "Metric", values_to = "Count")
```

```
# Plot multiple bars side-by-side
```

```
ggplot(temp_long, aes(x = SUBREGION, y = Count, fill = Metric)) +
  geom_bar(stat = "identity", position = "dodge") +
  scale_fill_manual(
    name = "",
    values = c("TOTAL_AFFECTED" = "steelblue", "TOTAL_DEATHS" = "red"),
    labels = c("Total Affected", "Total Deaths")
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    title = "Impact of 2004 Indian Ocean Earthquake and Tsunami by Subregion",
    x = "Subregion",
    y = "No. of Persons"
  )
)
```



Impact on Southern Asia

- Sri Lanka, which is an island nation, has the highest count of most affected people. Sri Lanka's both western and eastern coasts were hit, the latter was hit first as the ripples of the tsunami came from the east. However, since the intensity of the ripples decreased as the waves travelled over the Indian ocean, the number of deaths recorded in Sri Lanka is way lower than its counterpart island nation Indonesia in South Eastern Asia.
- India, which is a peninsula, suffered on both coasts, eastern and western, the former hit first. Island group of Andaman and Nicobar was severely hit as it was just 250 km away from the epicenter, followed by the state of Tamil Nadu on the South Eastern coast of the peninsula. Since, it is not an island, the

number of death was significantly lower than the Sri Lankan nation.

Impact on South Eastern Asia

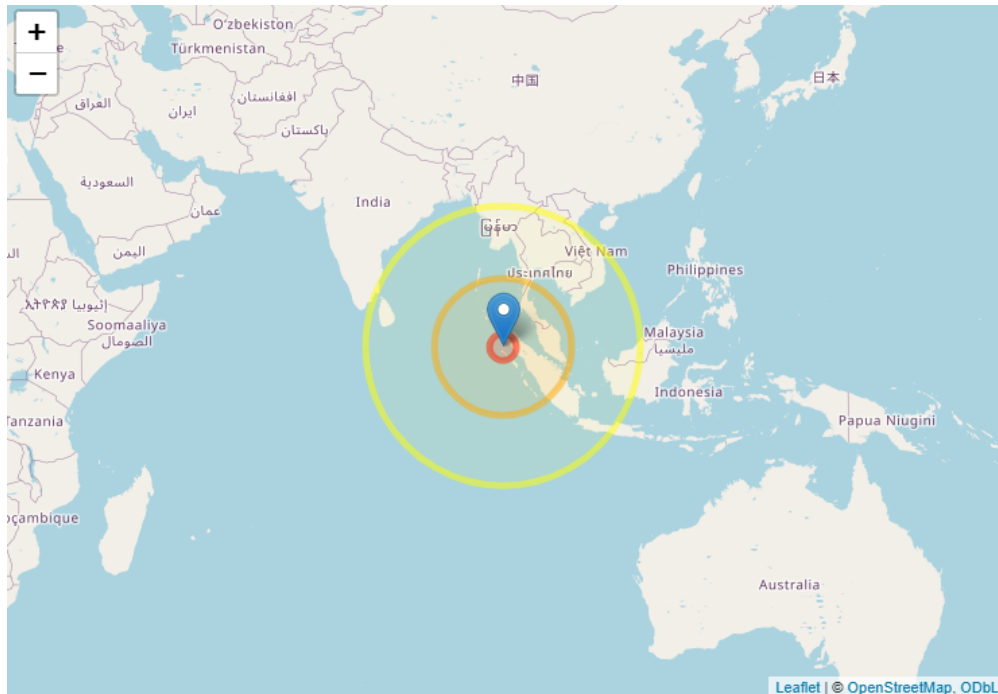
- Indonesia is the most severely hit country in the disaster with most affected people and most deaths occurring. The reasons are its close proximity to the epicenter (only 150-200 km) and its geography as an island nation.
- Thailand, not a primarily island nation, hosts tons of tourists on its islands. Hence, when the disaster occurred, it hit the tourists the most, and hence, the island regions of Thailand such as Phuket were most impacted and recorded the most deaths.

Impact on Sub-Saharan Africa

The 2004 tsunami waves traveled over 5,000–6,000 km across the Indian Ocean, eventually reaching the eastern coasts of Africa 7–12 hours after the earthquake. Somalia was the worst affected in Sub-Saharan Africa due to its direct exposure to the open ocean, lack of natural barriers, and absence of early warning systems, resulting in nearly 300 deaths and over 100,000 people affected.

The epicenter of the earthquake can be seen in the following map visual:

```
#earthquake_center <- tibble(  
#   name = "Indian Ocean Earthquake and Tsunami Epicenter",  
#   lon = unique(temp_df$LONGITUDE),  
#   lat = unique(temp_df$LATITUDE)  
#)  
#leaflet() %>%  
#   addTiles() %>%  
#   setView(lng = earthquake_center$lon, lat = earthquake_center$lat, zoom = 3) %>%  
#   addMarkers(  
#     lng = earthquake_center$lon,  
#     lat = earthquake_center$lat,  
#     popup = earthquake_center$name  
#   ) %>%  
#   addCircles(lng = earthquake_center$lon, lat = earthquake_center$lat,  
#               radius = 200000, color = "red", fillOpacity = 0.15) %>%  
#   addCircles(lng = earthquake_center$lon, lat = earthquake_center$lat,  
#               radius = 1000000, color = "orange", fillOpacity = 0.1) %>%  
#   addCircles(lng = earthquake_center$lon, lat = earthquake_center$lat,  
#               radius = 2000000, color = "yellow", fillOpacity = 0.05)
```



2004 Indian Ocean Earthquake Epicenter

(Kindly note that this image is created using the code written above. Since, HTML outputs are not allowed in Non-HTML formats, attaching as screenshot of the generated visual)

To read more about the disaster, kindly visit: [Indian Ocean Earthquake and Tsunami](#)

5. Outcomes Achieved

- **Identified the Most Frequently Occurring Disaster and Its Geographic Prevalence**
Floods were found to be the most frequent disaster type, with **4,140 recorded instances**, predominantly affecting **South-Eastern Asia (604 floods)** and **Southern Asia (172 floods)**.
- **Determined the Disaster with Longest Median Duration and Its Hotspots**
Droughts had the longest median duration of **274 days**, with major impacts in **Sub-Saharan Africa (432.8 million affected)** and **Asia (1.17 billion affected)**.
Specifically, **Eastern Asia** and **Southern Asia** were the most affected subregions, including two major droughts in **India**.
- **Identified the Most Impacted Region and Subregion Globally and the Prominent Disasters**
Asia was the most affected region overall (40.3% of disasters), while **Sub-Saharan Africa** was the most impacted subregion with **3,531 disasters**.
The most prominent disasters in Sub-Saharan Africa include *Floods*, *Road Accidents*, and *Epidemics*.
- **Classified the Most Fatal Disasters by Type and Their Locations**
Earthquakes accounted for the largest number of deaths (nearly 44.2% of all disaster-related deaths), followed by *Extreme Temperatures*, *Storms*, *Floods*, and *Epidemics*.
Most fatal earthquake events occurred in **Haiti (2010)**, **Indonesia (2004)**, **China**, **Pakistan**, and **Turkiye**.
- **Analyzed Most Fatal Disasters by Year**
The years **2010**, **2004**, and **2008** were the deadliest, primarily due to:
 - The **Haiti Earthquake (2010)**
 - The **Indian Ocean Earthquake and Tsunami (2004)**
 - The **Myanmar Cyclone (2008)**
- **Conducted Dedicated Case Studies for High-Impact Scenarios**
 - A **regional case study for India** analyzed the country’s high-impact *Floods*, *Droughts*, and *Earthquakes* over the years, showing major humanitarian effects in **2001**, **2004**, and **2005**.
 - A **disaster-specific case study of the 2004 Indian Ocean Earthquake and Tsunami** detailed its cross-continental impact across **South-Eastern Asia**, **Southern Asia**, and **Eastern Africa**.
- **Identified Disasters Associated with a Single Region Only**
Epidemics (especially bacterial and viral) were found to be highly concentrated in **Sub-Saharan Africa**, with minimal occurrences elsewhere — suggesting region-specific vulnerability and transmission dynamics.

Possible Avenues to Explore

- **Temporal Analysis:** Segment disasters into three periods—2000–2005, 2006–2010, and post-2010—to compare trends in frequency, impact, and response.
 - **Aid Prediction Model:** Fit a classification model to predict whether aid was received, using features like region, disaster type, severity, and people affected.
 - **Geopolitical Bias:** Explore if countries like the US show regional or strategic bias in their aid disbursement patterns.
-

6. Results and Discussions

A. One-Dimensional Summaries

- **Disaster Classification**

- *Floods* dominate the dataset with **4,140 instances**, followed by *Storms* (**2,677**) and *Road accidents* (**2,188**).
- *Droughts* have the **longest median duration** at **274 days**, followed by *Epidemics* (**92.5 days**).
- Notably, biological and climatological disasters have extremely long tail durations, indicating chronic impact.

- **Region-Wise Impact**

- **Asia** accounts for **40.3%** of all disasters, followed by **Africa (25.8%)**, **Americas (19.9%)**, and **Latin America and the Caribbean** being the most affected subregion within the Americas.
- Within Asia, **Southern Asia (2,158)** and **Eastern Asia (1,848)** are the most affected.
- **Sub-Saharan Africa** alone reports **3,531** disasters — the highest among subregions.

- **Temporal Trends (2000–2025)**

- Highest frequency of disasters occurred between **2000–2005**.
 - Events declined between 2006–2010 and stabilized post-2011.
 - Analysis suggests three temporal clusters:
 1. Rapid onset and high frequency (2000–2005)
 2. Gradual decline (2006–2010)
 3. Stabilized intensity and spread (2011 onwards)
-

B. Two-Dimensional Summaries

B.1 Disaster Classification × Region

- **Droughts - Disaster with Longest Median Duration**

- Severely impacted **Sub-Saharan Africa** with **432.8 million people affected**, and **Asia** with **1.17 billion**.
- Within Asia, **Eastern Asia** alone saw **359 million affected**, and **Southern Asia** reports **735 million affected**.
- In India, droughts in **2002** and **2015** affected more than **630 million people**, ranking among the most impactful events globally.
- Sub-Saharan Africa alone experienced **172 droughts** out of 176 droughts in the African region, with 43 of them receiving international aid responses.

- **Epidemics** - Disaster with 2nd Longest Median Duration
 - Heavily concentrated in **Sub-Saharan Africa** with **569 instances**, including **349 bacterial** and **169 viral** epidemics.
 - Region recorded **5.9 million people affected** and **37k deaths due to viral infections** and **52k deaths due to bacterial infections** by epidemics.
 - OFDA/BHA responded to **26 epidemic cases** in Africa.
- **Floods** - Disaster with Highest Frequency
 - Most frequent disaster type, significantly impacting **South-Eastern Asia (604 floods)** and **Southern Asia (172 floods)**.
 - These regions collectively experienced the highest exposure to flood-related devastation.

B.2 Disaster Start Year × Region

- **Asia** remained the most disaster-prone region every year, especially Southern and South-Eastern subregions in the early 2000s.
- **Africa**, primarily **Sub-Saharan**, experienced consistent disasters throughout the period — though some improvement was seen post-2015.
- **Latin America and the Caribbean** showed elevated disaster incidence until around 2010.
- **Eastern Asia**, while initially high-impact, showed decreasing trends after 2010, indicating strengthened resilience and preparedness measures.

C. Top-N Deep Dives

C.1 Total Deaths by Disaster Classification

Earthquakes alone account for nearly half of all recorded disaster-related deaths, followed by extreme temperature, storms, floods, and epidemics collectively contributing the majority of the remaining share.

Notable Earthquake Events

- **2010 Haiti Earthquake:** 222,750 deaths
- **2004 Indian Ocean Earthquake:** 165,708 deaths in Indonesia alone out of ~226k deaths
- **2008 Myanmar Cyclone:** Over 130,000 deaths

C.2 Total Deaths by Year

The years **2010**, **2004**, and **2008** recorded the highest death tolls due to catastrophic earthquakes, tsunamis, and cyclones respectively.

2012, despite “**End of the World**” myths, recorded **17,694 deaths** — not the worst, but still tragic.

D. Case Studies

D.1 Disasters in India

- **Floods**
 - 199 events, causing **35,270 deaths** and affecting **183 million people**.
 - Major events in **2000, 2002, and 2004** in the **Ganges river basin**, all due to torrential monsoon rains.
- **Droughts**
 - 2002 and 2015 events each affected more than **300 million**.
 - The 2015 drought ran for around **2 years**.
 - Represent India's most severe humanitarian crises, despite low reported death counts.
- **Earthquakes**
 - **2001 Gujarat Earthquake**: Magnitude 7.7, **20,000+ deaths**.
 - **2004 Indian Ocean Quake**: Magnitude 9.1, **16,389 deaths** in India alone.
- **Yearly Trend (India)**:
 - Death toll peaked in **2001 and 2004** due to the Gujarat earthquake and Indian Ocean tsunami respectively.
 - Disaster frequency peaked in **2005** due to **17 floods and 12 road accidents**.

D.2 2004 Indian Ocean Earthquake and Tsunami

- **Magnitude**: 9.1
- **Date**: December 26, 2004
- **Total Deaths**: 226,408
- **Affected People**: 2.4 million across 12 countries

Regional Breakdown:

- **Indonesia**: An island nation, first hit (165,708 deaths) due to proximity to epicenter
 - **Sri Lanka**: An island nation with 35,399 deaths due to double-coast impact
 - **India**: 16,389 deaths, mainly in Tamil Nadu and Andaman & Nicobar Islands
 - **Thailand**: Not a primary island nation, but has vacation islands such as Phuket, recorded '8.3k deaths
 - **Somalia**: Affected after 7–12 hours, showing tsunami's far-reaching effects
-

E. Regional Deep Dives

E.1 Sub-Saharan Africa

- **3,541 total disasters** — highest of all subregions
- Floods (**884**), epidemics (**569**), and road accidents (**758**) dominate the count

- **Poverty rates** range between **30%–80%**, correlating with disaster vulnerability and lack of response systems
- Sub-Saharan Africa seems to have done better over the years to reduce the epidemic instances. Similar inference can be attached to the road accidents with the help of improved infrastructure. Though the numbers for Storms and Water-based accidents seem to be low, the region is constantly affected by these over the years.
 - This is the most poverty-stricken region of the world

E.2 China

- **Most flood-affected country** in Eastern Asia
- Out of **305 floods** and **950 million affected** in the subregion, **China accounts for 221 floods** and **943 million affected people**
- Peak in **2006**: 20 flood events causing **430 deaths** and impacting **14 million people**, all linked to **Yangtze River Basin**

END OF THE REPORT
Predictive Analytics to follow in Semester 2
